



Discovery and biosynthesis of cyclic plant peptides via autocatalytic cyclases

Desnor N. Chigumba^{1,6}, Lisa S. Mydy^{1,6}, Floris de Waal^{2,6}, Wenjie Li¹, Khadija Shafiq¹, Jesse W. Wotring¹, Osama G. Mohamed^{1,3,4}, Tim Mladenovic¹, Ashootosh Tripathi^{1,3}, Jonathan Z. Sexton^{1,5}, Satria Kautsar^{1,2}, Marnix H. Medema²✉ and Roland D. Kersten¹✉

Many bioactive plant cyclic peptides form side-chain-derived macrocycles. Lyciumins, cyclic plant peptides with tryptophan macrocyclizations, are ribosomal peptides (RiPPs) originating from repetitive core peptide motifs in precursor peptides with plant-specific BURP (BNM2, USP, RD22 and PG1beta) domains, but the biosynthetic mechanism for their formation has remained unknown. Here, we characterize precursor-peptide BURP domains as copper-dependent autocatalytic peptide cyclases and use a combination of tandem mass spectrometry-based metabolomics and plant genomics to systematically discover five BURP-domain-derived plant RiPP classes, with mono- and bicyclic structures formed via tryptophans and tyrosines, from botanical collections. As BURP-domain cyclases are scaffold-generating enzymes in plant specialized metabolism that are physically connected to their substrates in the same polypeptide, we introduce a bioinformatic method to mine plant genomes for precursor-peptide-encoding genes by detection of repetitive substrate domains and known core peptide features. Our study sets the stage for chemical, biosynthetic and biological exploration of plant RiPP natural products from BURP-domain cyclases.

Plants produce many natural peptide products with diverse macrocyclization chemistry and biological activities¹. Cyclic plant peptides such as cyclotides, which are roughly 30 amino-acid-long peptides defined by macrocyclization via three disulfide bonds and head-to-tail cyclization, show remarkable stability, cell membrane penetration and diversifiability. Thus, they constitute lead structures for the development of pesticides and drug candidates for cancer treatment and antiviral therapy². Besides cyclization via disulfide bonds and ligation of peptide termini, plants are the source of multiple bioactive cyclic peptides with chemically diverse side-chain macrocyclizations via tryptophans and tyrosines (Fig. 1a, and Extended Data Fig. 1).

The genetic and biochemical basis of plant peptide macrocyclization is only characterized for head-to-tail-cyclic plant peptides, which are ribosomally encoded and posttranslationally modified peptide natural products (RiPPs) defined by a final protease-catalyzed macrocyclization^{2–9}. The genetic basis of side-chain-macrocylic plant peptides is largely unknown, except for the recent characterization of a lyciumin precursor gene from Chinese wolfberry encoding a C-terminal BURP (BNM2, USP, RD22 and PG1beta) domain (Pfam 03181)¹⁰ and an N-terminal domain with multiple lyciumin core peptides¹¹. BURP-domain proteins are plant proteins with a conserved CHX₁₀CHX_{25–27}CHX_{25–26}CH sequence motif identified in their founding members BNM2, a microspore-derived embryo protein from *Brassica napus*¹², USP, an unidentified seed protein from *Vicia faba*¹³, RD22, a drought-responsive protein from *Arabidopsis thaliana*¹⁴ and PG1β, the β-subunit of polygalacturonase 1 involved in fruit ripening in *Solanum lycopersicum*¹⁵. Despite their diverse roles in plant physiology, the biochemistry of BURP domains is uncharacterized¹⁰. Ribosomal biosynthesis of lyciumins, and the absence of bacterial-type or fungal-type multi-modular nonribosomal peptide synthetase genes from published plant genomes,

suggest a ribosomal origin of other side-chain-macrocylic plant peptides, potentially via BURP-domain precursor peptides. In recent years, multiple bacterial and fungal RiPP classes have been discovered, which are defined by side-chain-macrocylic bonds including tryptophans and tyrosines similar to their plant-derived counterparts^{16–24} (Extended Data Fig. 1). The macrocyclization in these bacterial RiPPs, such as streptide and darobactin, is catalyzed by radical S-adenosyl-L-methionine- (SAM-) dependent Fe-S-cluster enzymes^{16,22}. In fungal RiPPs, such as asperipin-2a and ustiloxins, it is dependent on DUF3328-containing oxidases²⁵. In addition, bacterial endosymbionts have been characterized as alternative producers of cyclic peptides isolated from plants²⁶. Identifying the genetic basis of undefined plant cyclic peptides could enable their gene-guided discovery from plant genomes, as highlighted by lyciumin discovery across angiosperms¹¹, and yield new biocatalysts for cyclic peptide engineering in vivo and in vitro^{6–8}.

Although plant peptides with side-chain macrocyclizations are intriguing in their chemical structures, biosynthesis and biological activities, their systematic discovery and biosynthetic characterization from plants are challenging for multiple reasons. First, plant metabolic extracts are complex and cyclic peptides are often low-abundance analytes in crude source plant extracts, which prevents their characterization by tandem mass spectrometry (MS/MS)-based metabolomics^{27–30}, as peptide parent ions are not selected for data-dependent MS/MS fragmentation or yield poor MS/MS fragmentation data due to low parent ion intensities. Second, the identification and structure elucidation of side-chain-macrocylic peptides from MS/MS spectra is difficult because the presence of one or two side-chain-derived macrocyclizations in a short peptide substantially reduces the possibility to infer sequence information from their tandem mass spectra, whereas head-to-tail-cyclic peptides can be sequenced by MS/MS^{31,32}. In addition, many plant

¹Department of Medicinal Chemistry, University of Michigan, Ann Arbor, MI, USA. ²Bioinformatics Group, Wageningen University, Wageningen, Netherlands. ³Natural Products Discovery Core, Life Sciences Institute, University of Michigan, Ann Arbor, MI, USA. ⁴Pharmacognosy Department, Faculty of Pharmacy, Cairo University, Cairo, Egypt. ⁵Department of Internal Medicine, University of Michigan, Ann Arbor, MI, USA. ⁶These authors contributed equally: Desnor N. Chigumba, Lisa S. Mydy, Floris de Waal. ✉e-mail: marnix.medema@wur.nl; rkersten@umich.edu

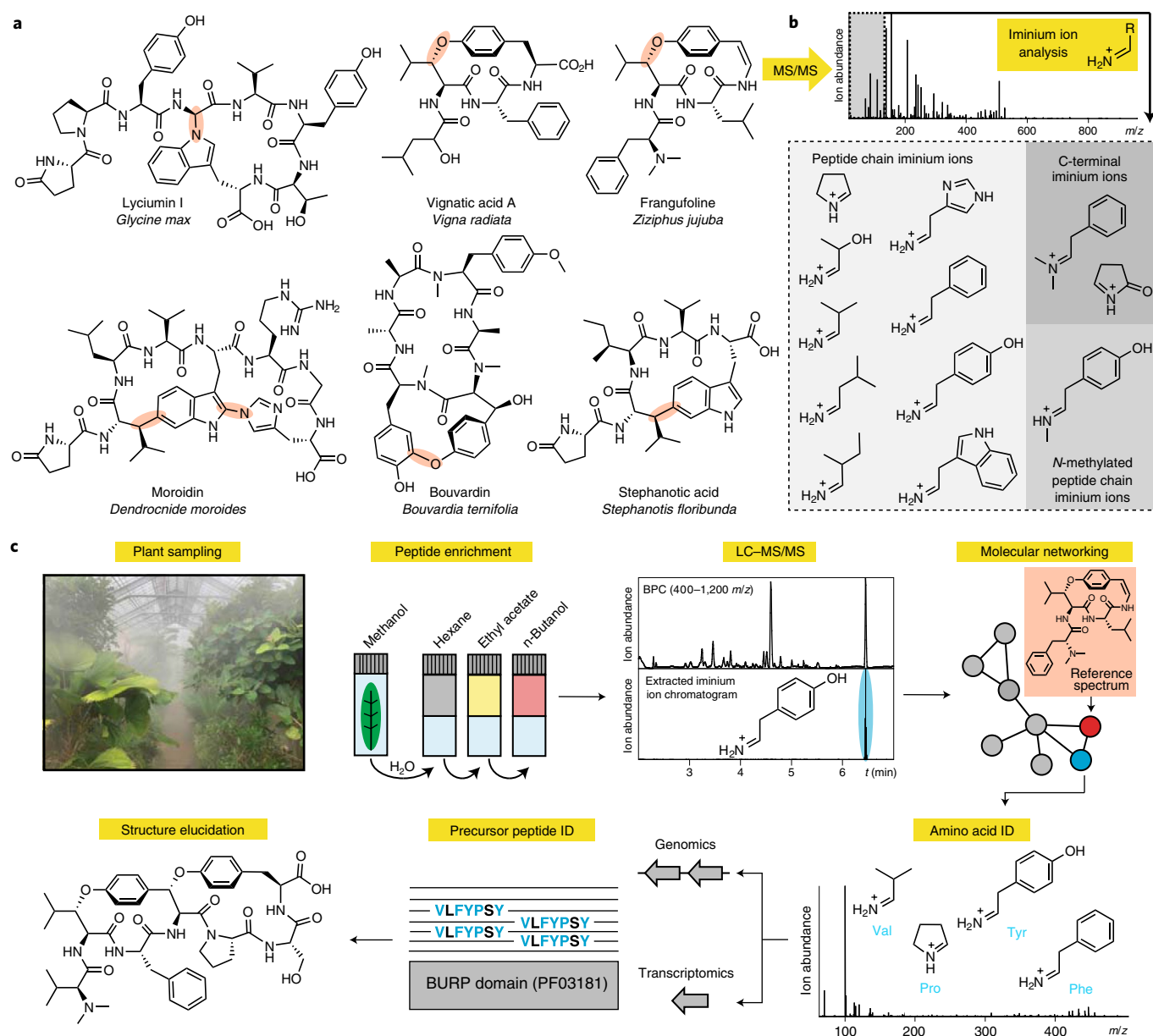


Fig. 1 | Systematic discovery approach of side-chain-macrocytic plant peptides. **a**, Representative structures of side-chain-macrocytic peptides isolated from plants with macrocyclic crosslinks highlighted in red. **b**, Detected iminium ions after HCD of representative side-chain-macrocytic peptides. **c**, Workflow for the systematic discovery of side-chain-macrocytic plant peptides.

peptide classes still do not have reference spectra in metabolomic databases, which can hinder their identification by metabolomic database searches such as Global Natural Product Social (GNPS) molecular networking^{29,31}. Herein, new metabolomic identification tools show great promise in systematic natural product classification from MS/MS data^{33,34}. Third, the genes involved in the biosynthesis of most side-chain-macrocytic peptides, that is precursor peptides, cyclases and proteases are largely unknown. This knowledge gap prevents the prediction of cyclic peptide chemistry, which could be linked to plant metabolomics data, from genes in plant genomes and transcriptomes.

Here, we address these bottlenecks in plant peptide discovery in a systematic approach to uncover new peptide macrocycles in plants by connecting peptidic analytes in botanical extracts to peptide biosynthetic genes bioinformatically identified in plant genomes and transcriptomes. Furthermore, our study presents a fundamentally

new family of copper-dependent peptide cyclases, which catalyze side-chain macrocyclizations in an autocatalytic manner.

Results

A cyclic plant peptide discovery workflow. We aimed to overcome the aforementioned problems in discovering side-chain-macrocytic plant peptides by (1) peptide-specific sample preparation for metabolomics analysis, (2) peptide-targeted metabolomic data acquisition and analysis and (3) connection of candidate peptide analytes in plant metabolomes with candidate precursor-peptide genes in plant genomes and plant transcriptomes.

To decrease metabolomic complexity, we partitioned crude methanol extracts as aqueous phases consecutively with hexane, ethyl acetate and *n*-butanol. Liquid chromatography–mass spectrometry (LC–MS) analysis of the organic solvent fractions showed that reference side-chain-macrocytic plant peptides such as lyciumins,

moroidin, stephanotic acid, deoxybouvardin and cyclopeptide alkaloids partitioned mainly into the ethyl acetate and *n*-butanol fractions (Extended Data Fig. 2). For peptide-targeted metabolomic data acquisition, we hypothesized that detection of amino-acid-specific iminium ions (Supplementary Table 1) could be a reliable way to identify MS/MS spectra of side-chain-macrocylic peptides and also enable characterization of amino acid content of a candidate peptide. To identify favorable MS fragmentation conditions for iminium ion detection, we analyzed reference plant peptides by high-energy collision-induced dissociation (HCD) on an Orbitrap at fragmentation energies of 10–40 eV (Supplementary Note 1). Our analysis showed that lyciumins, moroidin, stephanotic acid, bouvardins and cyclopeptide alkaloids yield at least one iminium ion mass signal among the 20 most intense fragment ions in their MS/MS spectra for fragmentation energies 25, 30 and 40 eV. The detected iminium ions were mostly derived from hydrophobic and aromatic amino acids and their *N*-methylated derivatives, as previously described³⁵ (Fig. 1b and Supplementary Note 1). To test whether cyclic plant peptides could be classified by the detection of an amino acid iminium ion, we analyzed known plant peptide structures with side-chain macrocyclizations for their amino acid content, and specifically for amino acids that are not crosslinked and are therefore detectable as iminium ions in MS/MS analysis. More than 98% of known side-chain-macrocylic plant peptides included at least one detectable amino acid (Supplementary Dataset 1), which should enable the identification of each peptide MS/MS spectrum by searching for proteinogenic amino acids.

Given our targeted MS detection of cyclic plant peptides, we envisioned the following workflow for cyclic peptide discovery from plants (Fig. 1c). First, plant samples were extracted by methanol, and the crude extracts were partitioned into hexane, ethyl acetate and *n*-butanol fractions. The *n*-butanol fractions were analyzed by LC–MS/MS with HCD at 25 eV to yield both MS/MS iminium ions for analyte identification and larger mass fragments for analyte structure elucidation. The metabolomic datasets were filtered for MS/MS spectra, which contained proteinogenic amino acid iminium ion masses among the 20 most intense mass signals, suggesting a peptidic parent ion. The filtered LC–MS/MS datasets were then subjected to GNPS molecular networking with reference MS/MS spectra of known side-chain-macrocylic plant peptides. Analytes in molecular networks were classified based on the presence of reference peptide spectra, and compound abundance in a given extract was estimated by precursor ion intensity. New peptide chemistry was predicted from nodes, that is MS/MS spectra, in spectral clusters with reference peptide spectra with unusual molecular weights and from nodes in clusters without reference peptide spectra. The tandem mass spectrum of a candidate cyclic peptide was then analyzed for the presence of proteinogenic amino acid iminium ions (Supplementary Table 1). Given the role of BURP-domain genes encoding side-chain-macrocylic plant RiPPs, BURP-domain genes were identified in the genome or transcriptome of the peptide source plant and the *N*-terminal domain of the BURP domain was searched for short sequences matching (1) the observed iminium ions and (2) the observed parent mass considering biosynthetic modifications such as cyclization and *N*-terminal protection, for example pyroglutamate formation and *N*-methylation. Finally, a match between a tandem mass spectrum of a candidate cyclic peptide and a candidate core peptide in a BURP-domain protein was used to inform chemical novelty and guide structure elucidation by nuclear magnetic resonance (NMR).

Discovery of cyclic peptides from botanical collections. We tested our peptide discovery workflow on plants from phylodiverse botanical collections. In total, 297 plant species representing 102 plant families at the University of Michigan Matthaei Botanical Garden and 51 legume specimens from the UM herbarium were analyzed

in their aerial tissues for cyclic peptides (Supplementary Dataset 2). We tested herbarium specimens in addition to live plants as natural product sources because herbaria have been described as underused resources for phytochemical exploration³⁶. Legumes were selected because they have been reported as producers of diverse cyclic peptide chemistry^{11,37,38}. The molecular network of iminium ion-filtered LC–MS/MS datasets comprised several spectral clusters with reference peptide mass spectra, including a cyclopeptide alkaloid cluster, three lyciumin clusters and a stephanotic acid cluster (Fig. 2a and Extended Data Fig. 3). With 453 nodes, the cyclopeptide alkaloid-containing cluster was the second-largest cluster in the peptide-filtered molecular network including MS/MS data from 261 plant species representing 92 plant families. The characterization of candidate cyclopeptide alkaloid analytes in metabolomes of phylodiverse source plants including lycophytes, ferns, gymnosperms, monocots and dicots indicates a much broader distribution of these phytochemicals in the plant kingdom than previously known (Supplementary Dataset 3). Among the most prolific cyclopeptide alkaloid sources in terms of chemical diversity measured by the number of detected nodes were, besides the known source plant New Jersey tea (*Ceanothus americanus*), cycads such as the Eastern Cape giant cycad (*Encephalartos altensteinii*) and arum plants such as the Philippine evergreen (*Aglaonema commutatum*, Extended Data Fig. 4).

To identify new peptide chemistry from the cyclopeptide alkaloid cluster, we focused on nodes with parent masses larger than the average cyclopeptide alkaloid mass of 562 Da (Supplementary Dataset 1). Several nodes from African clubmoss (*Selaginella kraussiana*) had parent masses above 900 Da, indicating larger structures than common cyclopeptide alkaloids. One of the most abundant candidate peptides in *S. kraussiana* extract was an analyte 1 with the monoisotopic mass of 912.445 Da (Fig. 2a). Applying our workflow, we could connect 1 to candidate core peptide VLFYPSY in a BURP domain, SkrBURP, from a *S. kraussiana* leaf transcriptome by considering peptide bicyclization and dimethylation (Fig. 2c, Extended Data Fig. 5 and Supplementary Fig. 1). The target peptide 1 named selanine A was purified from *S. kraussiana* and its NMR analysis revealed a bicyclic peptide matching the predicted core peptide VLFYPSY with a *N,N*-dimethylated valine and a dityrosine-ether-crosslink between the leucine-2, tyrosine-4 and tyrosine-7 (Fig. 2b, Supplementary Fig. 1, Supplementary Table 2 and Supplementary Notes 2 and 3). A monocyclic cyclopeptide alkaloid 2, named selanine B, from *S. kraussiana* corresponding to a node with the parent mass 567.128 Da, was characterized as the VLFY ring of selanine A (Fig. 2b, Supplementary Fig. 2, Supplementary Tables 3 and 4 and Supplementary Notes 2 and 4), indicating that both mono- and bicyclic cyclopeptide alkaloids could be derived from the candidate BURP-domain precursor SkrBURP. Bicyclic cyclopeptide alkaloids are unprecedented in plants, but a natural product with a similar dityrosine-ether-crosslink called asperipin-2a (Extended Data Fig. 1) has been isolated from the fungus *Aspergillus flavus*¹⁸. Another abundant candidate peptide above the cyclopeptide alkaloid average mass was analyte 3 with the parent mass 713.376 Da and MS/MS data corresponding to a glycosylated cyclopeptide alkaloid (Supplementary Fig. 5) from an *E. altensteinii* leaf extract (Fig. 2a). Isolation and structure elucidation of 3, named encephanine, revealed a 14-membered cyclopeptide alkaloid with a β -D-glucose attached to an ortho-hydroxyl-group of a *p,o*-dihydroxystyrylamine moiety (Fig. 2b, Supplementary Fig. 3, Supplementary Table 5 and Supplementary Notes 2 and 5). As no transcriptome of *E. altensteinii* was available at the time of this study, the identification of encephanine, the first reported glycosylated cyclopeptide alkaloid, shows that new cyclic peptide chemistry can also be discovered by our approach without genetic information.

The stephanotic acid-containing cluster included analytes from Eastern redbud (*Cercis canadensis*). As the Eastern redbud genome

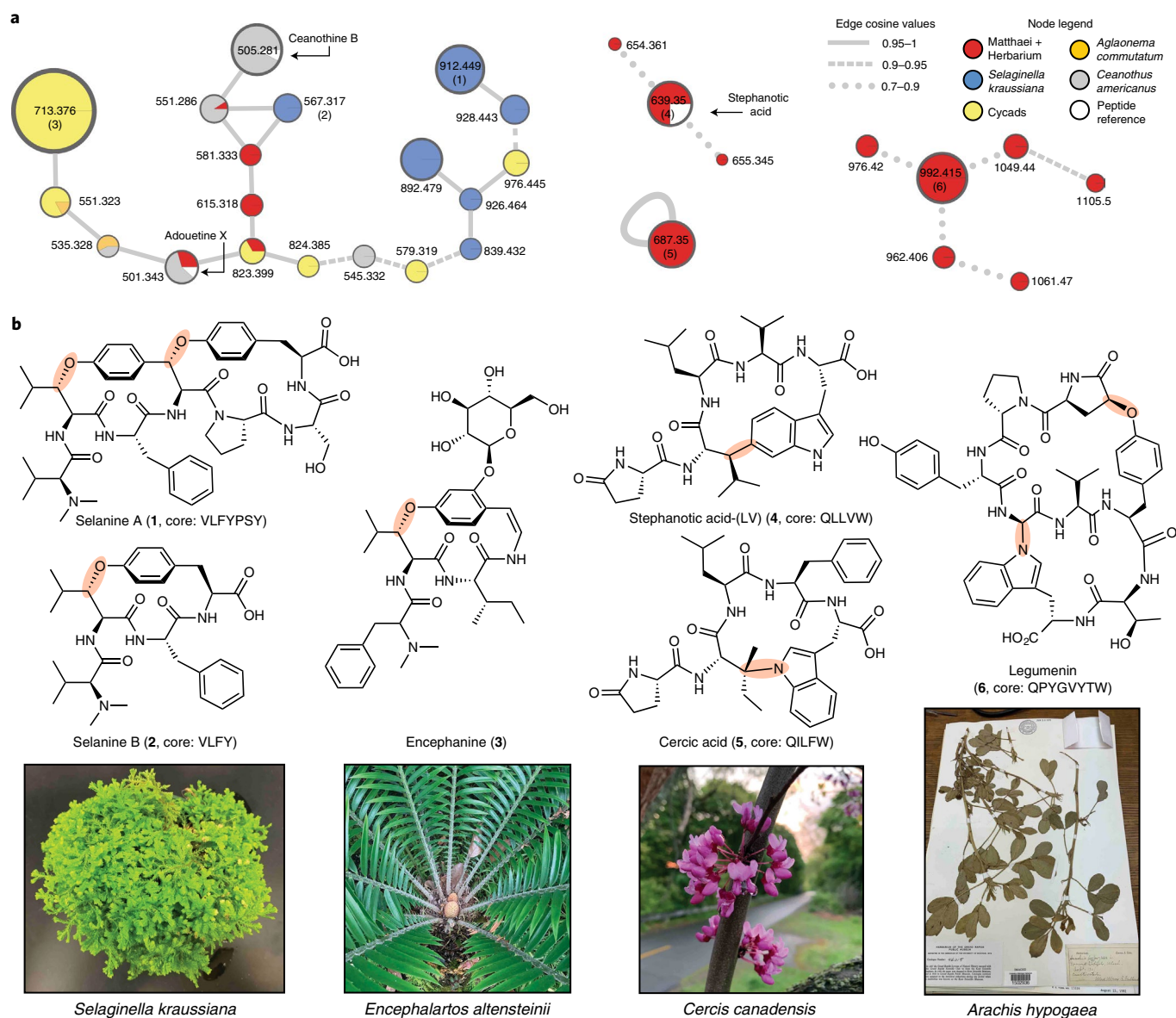


Fig. 2 | Discovery of side-chain-macrocytic peptides in botanical collections. **a**, Spectral clusters of peptide-filtered metabolomic datasets from botanical collection samples. Node sizes represent relative precursor ion intensity. Pie charts represent relative spectral counts of defined groups. **b**, Structures of characterized cyclic peptides from botanical collections. Macrocytic bonds are highlighted in red. Abbreviations: *N,N*-Me₂Phe, *N,N*-dimethylated phenylalanine; *N,N*-Me₂Val, *N,N*-dimethylated valine and PyroGlu, pyroglutamate.

was recently sequenced³⁹, we focused on analysis of candidate peptides in its metabolome. *Cercis* analyte 4 from the stephanotic acid cluster matched in its MS and MS/MS data to a BURP-domain protein, CcaBURP2, with N-terminal repeats including candidate core peptide QLLVW (Extended Data Fig. 5 and Supplementary Fig. 4). Isolation and structure elucidation of 4 revealed a stephanotic acid-type peptide, stephanotic acid-LV, from leaf and stem extracts of Eastern redbud trees (Fig. 2b, Supplementary Fig. 4, Supplementary Tables 6 and 7 and Supplementary Notes 2, 6 and 7). Another abundant peptidic analyte 5 in the extract of *C. canadensis*, which did not cluster with any peptide from the botanical garden samples, connected to BURP-domain CcaBURP1 from the *C. canadensis* genome with repetitive candidate core peptide QILFW (Extended Data Fig. 5 and Supplementary Fig. 5). NMR analysis of 5, named cercic acid, showed a C–N cross-link between the β -carbon of isoleucine and the indole-nitrogen of the C-terminal tryptophan (Fig. 2b, Supplementary Fig. 5,

Supplementary Table 8 and Supplementary Notes 2 and 8). In addition to spectral clusters containing reference peptide spectra, we investigated clusters without annotations, because they might contain new peptide macrocycles. One cluster of our botanical metabolomes was enriched in analytes from herbarium specimens and included a central analyte 6, which was present in five different legume species, including peanut (*Arachis hypogaea*). Based on its MS data, 6 corresponded to core peptide QPYGVYTW in two BURP-domain proteins in the peanut genome⁴⁰ after consideration of pyroglutamate formation and bicyclization (Extended Data Fig. 5 and Supplementary Fig. 6). We tested several legumes for 6 and isolated it from alfalfa (*Medicago sativa*) seed extracts (Fig. 3). NMR analysis of 6, named legumenin, revealed a bicyclic structure corresponding to the predicted core peptide with a lyciumin-type C–N crosslink between the glycine-4- α -carbon and the C-terminal tryptophan-indole-nitrogen and a C–O crosslink between the tyrosine-6-hydroxyl-group and the γ -carbon of the N-terminal

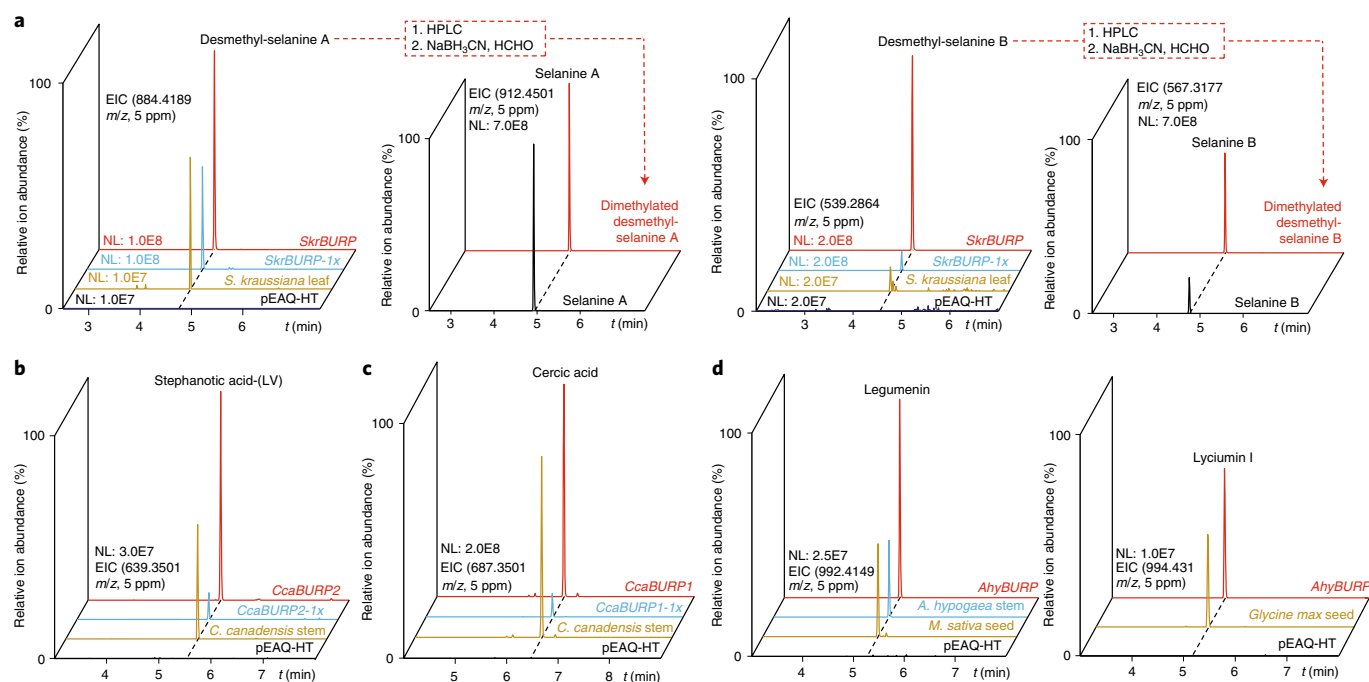


Fig. 3 | Transient expression of BURP-domain precursor peptides in *N. benthamiana*. **a**, Detection of desmethyl-selanine A and B in *N. benthamiana* leaves after 6 days of transient expression of *SkrBURP* or *SkrBURP*-[1xVLFYPSY] in transient expression vector pEAQ-HT, and detection of selanine A and B after reductive dimethylation of HPLC-purified desmethyl-selanine A and B, respectively. **b**, Detection of stephanotic acid-(LV) in *N. benthamiana* leaves after 6 days of transient expression of *CcaBURP2* and *CcaBURP2*-[1xQLLVW] (*CcaBURP2*-1x) in pEAQ-HT. **c**, Detection of cercic acid in *N. benthamiana* leaves after 6 days of transient expression of *CcaBURP1* and *CcaBURP1*-[1xQLFW] (*CcaBURP1*-1x) in pEAQ-HT. **d**, Detection of legumenin and lyciumin I (Fig. 1a) in *N. benthamiana* leaves after 6 days of transient expression of *AhyBURP* in pEAQ-HT.

pyroglutamate (Fig. 2b, Supplementary Fig. 6, Supplementary Table 9 and Supplementary Notes 2 and 9).

We tested the new cyclic peptides for bioactivities. When tested against selected pathogenic microbes as well as cancer cell lines, no significant antibacterial and anticancer activity could be detected for 1–6 (Extended Data Fig. 6). Beyond medicinal bioactivities, a BLAST search of the selanine precursor *SkrBURP* yielded two homologous proteins from mung bean (*Vigna radiata*), which are associated with a bruchid resistance locus in mung bean accession TC1966 (Extended Data Fig. 6)³⁸. A cyclopeptide alkaloid called vignatic acid A has been isolated from *V. radiata* TC1966 (ref. ³⁸) and characterized as an insecticide against azuki bean weevils. As the *SkrBURP*-homologous resistance genes from this strain are associated with vignatic acid production and include the core peptide LLFY matching the vignatic acid A structure, it can be hypothesized that similar cyclopeptide alkaloids such as selanine B also have insecticidal activity. Our metabolomic analysis of botanical collections yielded several peptides with new macrocyclization chemistry, via tyrosine- and tryptophan-side chains, raising the question of how they are biosynthesized.

Precursor BURP domains are autocatalytic peptide cyclases.

Given the diversity of cyclic peptide structures discovered from plant metabolomes by our approach, we set out to characterize the genetic and biochemical basis of plant peptide macrocyclizations involving tryptophan and tyrosine. We cloned candidate BURP-domain precursor peptides identified during peptide discovery from source plants and expressed them transiently in *Nicotiana benthamiana* with the pEAQ-HT expression vector via *Agrobacterium tumefaciens*-mediated infiltration⁴¹. Metabolomic analysis of transgenic tobacco leaves showed that *SkrBURP* expression resulted in the production of desmethylated analogs of selanine A and B, which could readily be converted to selanine A and B, respectively,

by reductive dimethylation⁴². *CcaBURP1* and *CcaBURP2* expression yielded cercic acid and stephanotic acid-(LV), respectively, and expression of one of the candidate legumenin BURP-domain precursor peptides from peanut, named *AhyBURP*, produced both legumenin and lyciumin I, which lacks the PyroGlu1-Tyr6-crosslink of legumenin⁴¹. All peptides derived from transiently expressed BURP domains were detected in source plant extracts (Fig. 3 and Supplementary Fig. 7). These tobacco expression results confirmed mono- and bicyclic cyclopeptide alkaloids, stephanotic acid, cercic acid and legumenin as new plant RiPP families derived from BURP-domain precursor peptides.

As BURP domains are biochemically undefined and directly connected to six plant RiPP classes with macrocyclizations involving tyrosine and tryptophan, we hypothesized that these domains might be peptide cyclases. To test this hypothesis, we heterologously expressed *AhyBURP* in *Escherichia coli*, purified (Supplementary Fig. 8) and tested it for cyclization activity by incubation with Cu(II), Co(II), Fe(II), Ni(II) and Zn(II) at pHs of 5, 6 and 7 based on a study of *AhyBURP*-homolog Sali3-2 from soybean, which bound to several divalent metal ions in an affinity chromatography experiment^{43,44}. Bottom-up proteomic analysis of the *AhyBURP* reaction mixtures resulted in the detection of tryptic peptide IPFAQPYGVYTWGK, comprising the legumenin core peptide and a loss of four hydrogens compared to the linear peptide, in *AhyBURP* reactions with Cu(II) (Fig. 4a and Supplementary Fig. 9). In addition, the same tryptic peptide with a loss of two hydrogens could be detected. MS/MS analysis showed that the four-hydrogen-mass loss is located in the core peptide sequence QPYGVYTW, which is in agreement with legumenin-macrocylation. Similarly, MS/MS analysis showed that the two-hydrogen-mass loss is located in the sequence GVYTW, which is in agreement with lyciumin-macrocylation. Finally, incubation of high-performance liquid chromatography (HPLC) -purified bicyclic peptide IPFAQPYGVYTWGK with an

aminopeptidase and a carboxypeptidase resulted in the formation of legumenin (Fig. 4b), which confirmed AhyBURP-autocatalyzed bicyclization.

We further set out to confirm BURP domains of selanines, cercic acid and stephanotic acid as peptide cyclases, which all contain multiple core peptides compared to AhyBURP (Extended Data Fig. 5). Corresponding BURP domains, SkrBURP, CcaBURP1 and CcaBURP2, were expressed as mono-core peptide constructs with maltose-binding protein (MBP) tags (Fig. 5 and Extended Data Fig. 7) to maximize soluble protein expression in *E. coli*. We verified that transient expression of truncated BURP domains in *N. benthamiana* yielded the corresponding product of the wild-type BURP domains (Fig. 3). MBP-SkrBURP-[1xVLFYPSY] incubation with 1 mM Cu(II) resulted in the proteomic detection of a tryptic peptide AVPASVEDPSQADADNVLFYPSYAAR as a minor analyte with a loss of four hydrogens and a major analyte with a loss of two hydrogens (Fig. 5 and Supplementary Fig. 10). The mass losses were localized by MS/MS to the core peptide sequences LFYPSYA and LFY, respectively (Supplementary Fig. 10). Incubation of MBP-CcaBURP2-[1xQLLVW] with 1 mM Cu(II) resulted in the detection of a tryptic peptide TICDDLNSQLLVWR with a loss of two hydrogens (Extended Data Fig. 7 and Supplementary Fig. 11). The corresponding tryptic peptide was modified covalently by buffer-derived β -mercaptoethanol at its cysteine (Extended Data Fig. 7 and Supplementary Fig. 11). The mass loss of two hydrogens was localized to the core peptide sequence LLVWR based on MS/MS (Supplementary Fig. 11). Finally, incubation of CcaBURP1-[1xQILFW] with 1 mM Cu(II) resulted in the detection of tryptic peptide AIYDDPNSQILFWR with a loss of two hydrogens (Extended Data Fig. 7 and Supplementary Fig. 12). MS/MS analysis confirmed this mass loss to be localized to the core peptide sequence ILFW (Supplementary Fig. 12). Incubation of HPLC-purified bicyclic tryptic peptide AVPASVEDPSQADADNVLFYPSYAAR of SkrBURP-[1xVLFYPSY] with aminopeptidase and carboxypeptidase yielded both desmethyl-selanine A and desmethyl-selanine B (Extended Data Fig. 7), confirming SkrBURP as selanine cyclase. Incubation of HPLC-purified monocyclic peptide AIYDDPNSQILFWR of CcaBURP1-[1xQILFW] with aminopeptidase and carboxypeptidase resulted in the formation of cercic acid (Extended Data Fig. 7), which verified CcaBURP1 as cercic acid cyclase. For CcaBURP2, no monocyclic tryptic core peptide could be purified in sufficient quantities for exopeptidase assays. However, the MS/MS data of the modified CcaBURP2 core peptide also supports macrocyclic modification rather than dehydrogenation modifications, which would result in linear peptide fragmentation patterns. The detected modifications were generated by proteins purified and incubated under aerobic conditions, which indicates that BURP domains are at least oxygen-tolerant enzymes for single-turnover catalysis. It can be concluded that BURP domains are peptide cyclases that catalyze crosslinking of core peptide residues within their protein sequence in a Cu(II)-dependent posttranslational modification event.

Automated genome mining of BURP-domain plant peptides.

The connection between BURP-domain peptide cyclases and their core peptide substrate within a single gene raises the possibility of gene-guided discovery of cyclic peptides from plant genomes. As the number of sequenced plant genomes is rapidly growing⁴⁵, we set out to implement BURP-domain-based genome mining of cyclic peptides into the plant genome mining platform plantiSMASH⁴⁶. In our plantiSMASH extension, BURP-domain-encoding loci are identified in a given plant genome based on the BURP-domain protein family (Pfam PF03181). Each BURP-domain-encoding locus is then searched for core peptides of known plant RiPPs to predict plant peptides such as cyclopeptide alkaloids, legumenins, lyciumins, cercic acids and stephanotic acids (Fig. 6a). In addition to plant peptide classes introduced in this study, a large number of BURP-derived

RiPPs with unknown modifications, and therefore undefined core peptide motifs, are likely hidden in BURP-domain gene loci in plant genomes. Because most BURP-domain precursor peptides have N-terminal domains with multiple core peptides in sequence repeats (Extended Data Fig. 6)¹¹, we developed an algorithm named RepeatFinder that searches BURP-domain gene loci for the presence of sequence repeats (Fig. 6a). The algorithm searches every gene in the region of the BURP-domain locus to not exclude any potential peptides. Within each of these genes, it identifies repeated *k*-mers to allow pinpointing genes that potentially encode precursors for cyclization and proteolytic cleavage to yield mature cyclic peptides. To test the plantiSMASH extension for BURP-domain-based peptide discovery, we analyzed 79 plant genomes from 34 plant families and six green algal genomes. While no BURP-domain-encoding genes were identified in green algal genomes, genomes of terrestrial plants and the marine angiosperm *Zostera marina* contained on average nine BURP-domain-encoding gene loci, with 26% of these loci containing more than one BURP-domain gene. In 25% of BURP-domain-encoding loci, RepeatFinder identified sequence repeats surrounding a BURP-domain gene (Supplementary Dataset 4).

As a benchmark, RepeatFinder successfully identified lyciumin precursors in the genomes of amaranth, potato and tomato by detection of repetitive core peptide domains around BURP-domain loci and in the beet and *Solanum pennellii* genomes by the known motif check for lyciumin core peptides (Supplementary Fig. 13). Next, we wanted to test whether new cyclic peptides could be predicted by RepeatFinder from plant genomes. Among RepeatFinder-hits, *GLYMA_04G180400*, a BURP-domain-encoding gene from the soybean genome⁴⁷, showed 12 sequence repeats of eight amino acids. Although these repeats did not include residues indicating BURP-domain-catalyzed cyclization such as tyrosine and tryptophan, the sequences adjacent to the detected repeats included a tyrosine and were distinct from core peptide motifs of known plant RiPPs (Fig. 6b). Transient expression of synthetic *GLYMA_04G180400* in *N. benthamiana* resulted in the detection of two candidate monocyclic peptides matching putative core peptides VPIFY and VPPIFY from this BURP-domain (Fig. 6c,d). The candidate cyclopeptides could be detected in methanolic extracts of soybean roots and cyclo-[VPPIFY] was purified from soybean roots and structurally characterized based on NMR and MS/MS analyses (Fig. 6d, Supplementary Fig. 13, Supplementary Table 10 and Supplementary Notes 2 and 10) to include an ether-crosslink between the β -carbon of Pro3 and the hydroxyl-group of Tyr7. The characterization of the *GLYMA_04G180400*-derived peptides from soybean highlights that plantiSMASH BURP-domain analysis can yield new plant RiPPs from genome-sequenced plants. Overall, our plantiSMASH analysis of 79 plant genomes based on RepeatFinder and known core peptide prediction projects that 12% of BURP-domain loci likely code for cyclic peptide chemotypes (Supplementary Dataset 4).

Discussion

Genomic discovery of plant natural products is challenging due to difficult prediction of complete biosynthetic pathways, prediction of product structures of predicted pathways and connection of such predicted structures to analytes in complex plant metabolomes⁴⁸. Our genome mining approach of BURP-domain-derived peptides overcomes these challenges because the BURP domain is a scaffold-generating enzyme fused to its substrate, which can be characterized in its biosynthetic building blocks by MS/MS. To connect a BURP-domain gene predicted in a plant genome to a cyclic peptide analyte, the complete identification of its metabolic pathway is not required, but only the identification of the BURP domain and its core peptide domain. On a genomic level, this requires the characterization of open reading frames of a BURP-domain gene in

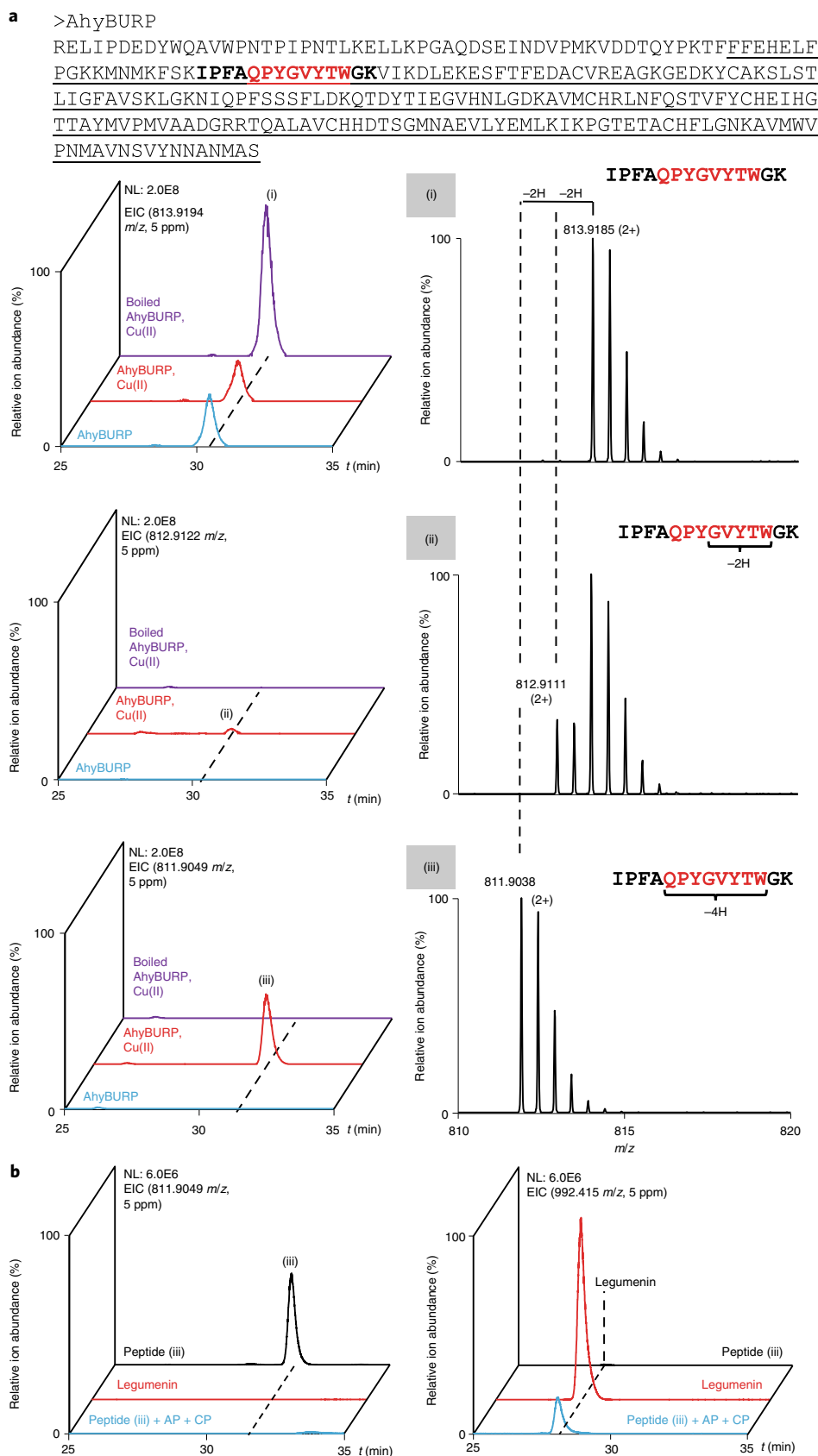


Fig. 4 | In vitro reconstitution of AhyBURP as an autocatalytic peptide cyclase. a, Detection of mono- and bicyclic core peptide modifications in purified AhyBURP after incubation with Cu(II) at pH 7. **b**, Detection of legumenin after incubation of modified (bicyclic) tryptic core peptide IPFAQPYGVYTWGK with aminopeptidase (AP) and carboxypeptidase (CP). BURP-domain sequences are underlined, core peptides are highlighted in red and bold sequences are characterized tryptic peptides including core peptides.

>SkrBURP-[1xVLFYPSY] (with N-terminal maltose binding protein)
 [MBP]-IERVHSAKEGVSEAMNDQGHPTIANK**AVPASVEDPSQADADN**VLFYPSY**AAR**PKAPSTMDHDA
 THNAAQNMMHHQKGALLFFRMNTLKVGGEVVIPS LASHALGGRKLMPRL EAMLGSMELPRLLSVLKIPL
 DSTLARKSKTNLGNCRSPPLQGEKKACVSSIRSMTNFARSVLEDKKPLEHLNPSRPVSSPEHFKIMDVK
 VVTENSVVTCHPMVFPYALYMCHFVPKSVPIKVTLODDHDKLVVVPVMCHMDTSEFDPSHLSFKILNTK
 PGAEAMCHWMPNISHIMWYTS DGKTRDVL

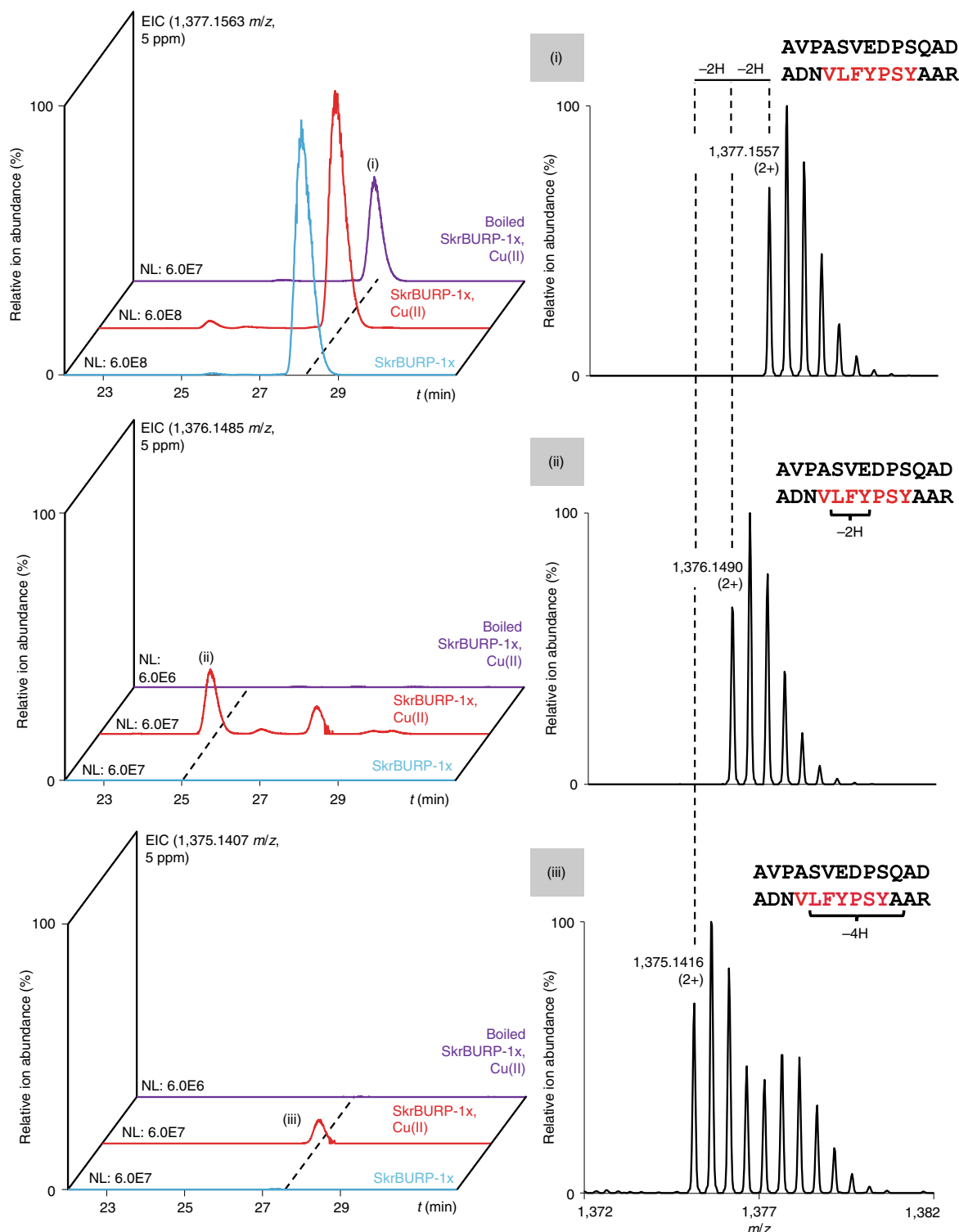


Fig. 5 | In vitro reconstitution of SkrBURP domain. Detection of mono- and bicyclic core peptide modifications in purified MBP-SkrBURP-[1xVLFYPSY] after incubation with Cu(II) at pH 7. BURP-domain sequence is underlined, core peptide is highlighted in red and bold sequence is characterized tryptic peptides including core peptide.

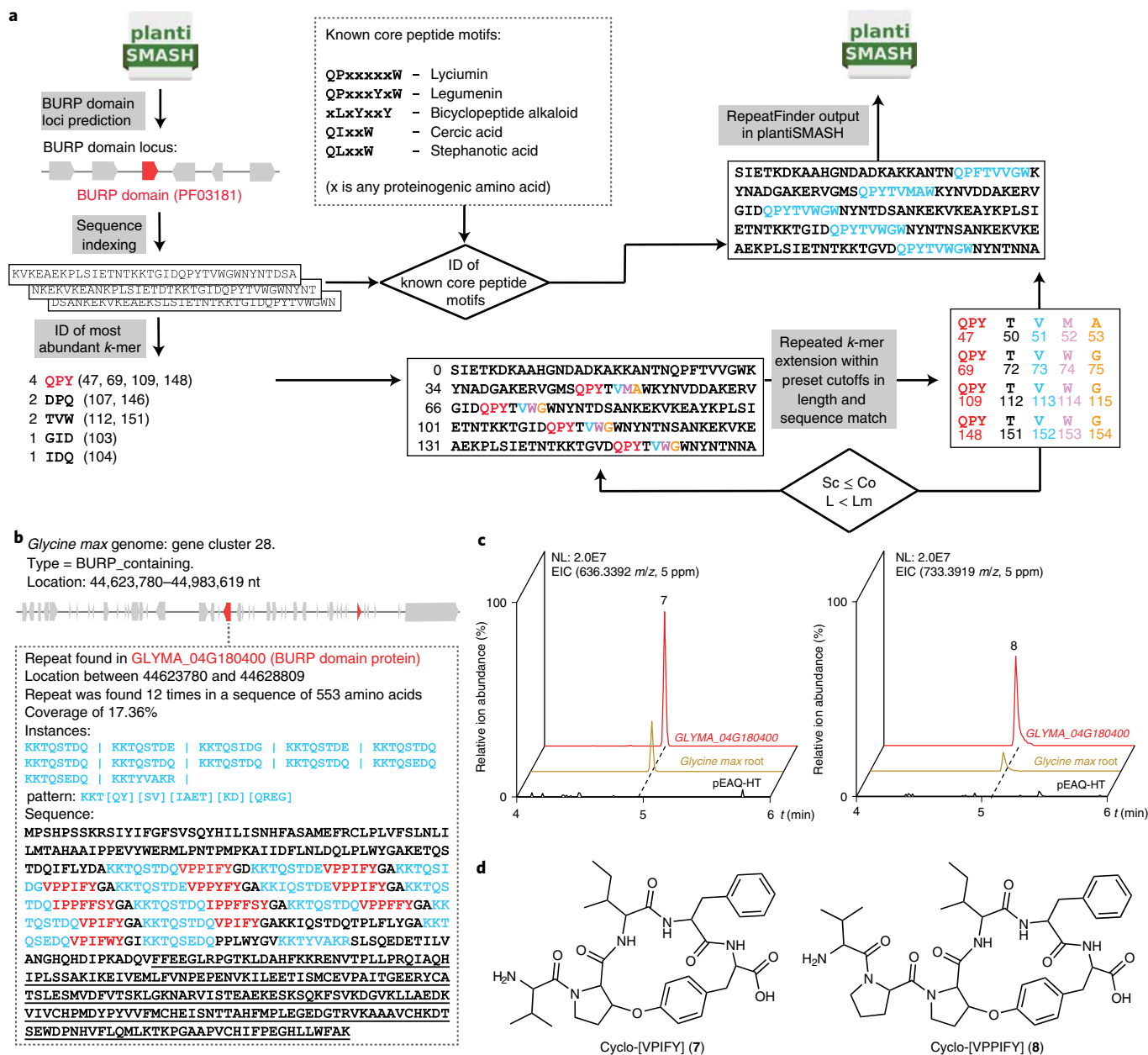


Fig. 6 | Genome mining of BURP-domain-derived peptides by plantiSMASH and RepeatFinder. **a**, RepeatFinder extension of plantiSMASH for genome mining of BURP-domain RiPPs based on known core peptide features and based on the detection of sequence repeats. **b**, Prediction of a BURP-domain precursor peptide from soybean genome by RepeatFinder in plantiSMASH. BURP-domain sequences are underlined, core peptides are highlighted in red and detected sequence repeats are highlighted in blue. **c**, Detection of candidate monocyclic peptides corresponding to two core peptides of GLYMA_04G180400 in soybean root extract and in *N. benthamiana* leaves after transient expression of GLYMA_04G180400 for 6 days. **d**, Predicted cyclopeptide structures derived from GLYMA_04G180400. Abbreviations: Sc, score; Co, cutoff; L, k-mer length and Lm, maximum k-mer length.

combination with transcriptomic data or Sanger sequencing. Our approach requires a mass spectrometer with MS/MS capabilities with a low-*m/z* cutoff in MS/MS spectra for the detection of amino acid iminium ions. For the characterization of a cyclic peptide in a plant extract, a genome of the source plant is preferred for the precursor peptide search but the peptide can also be discovered with a transcriptome or even without any genetic data. The strength of our approach is that unknown cyclic peptide chemistry can be accessed due to untargeted peptide analysis and due to the diversity of crosslinks formed between amino acids by BURP-domain cyclases. Limitations of our workflow are the occurrence of false positives during metabolomics analysis because of analytes, which

have fragment masses matching target amino acid iminium ions. Our approach still requires manual MS analysis of candidate cyclic peptides, especially of unannotated molecular clusters or singletons, which can harbor unknown cyclic peptides. The connection of a candidate cyclic peptide to a BURP-domain precursor gene can be undermined by incomplete or missing gene sequences due to low-quality assemblies and annotations of target genomes or transcriptomes. Finally, bioinformatic prediction of BURP-domain precursor peptides can also yield false positives based on known core peptide motifs predicted by RepeatFinder even in the absence of repeats. This could be reduced by adding repeat detection as an additional requirement for precursor peptide identification.

Automation in structural prediction of peptide analytes and their connection to predicted precursor peptides will streamline our workflow.

The key step in connecting a peptide analyte to a BURP-domain gene was the identification of a candidate core peptide sequence in a BURP-domain protein, which matches the observed amino acid iminium ions and the observed analyte mass after consideration of core peptide modifications such as cyclization and N-terminal protection. Based on plant RiPPs introduced in this and an earlier study¹¹, core peptides in BURP-domain proteins usually have a C-terminal tyrosine or tryptophan. A common N-terminal residue in plant RiPPs is glutamine-derived pyroglutamate, which can enable N-terminal core peptide definition. The prediction of core peptides can be guided by identification of sequence repeats in BURP domains. The RepeatFinder algorithm detects such sequence repeats without a priori knowledge of RiPP core peptide features and, thus, it can discover plant peptides with new macrocyclization chemistry.

Our study yields an updated biosynthetic proposal of BURP-domain-derived cyclic peptides¹¹ (Extended Data Figs. 8 and 9). In the first step, a BURP-domain precursor peptide is formed by the ribosome. Next, the BURP domain catalyzes macrocyclization modifications in its core peptide sequence after Cu(II) binding. The sequence of cyclization in bicyclic peptides, such as legumenin and selanine A, can be either C- to N-terminal or N- to C-terminal cyclization, respectively. The modified core peptide is then proteolytically released from the BURP-domain precursor, with an N-terminal protection modification following the N-terminal proteolytic cleavage of the core peptide. N-terminal pyroglutamate formation seems to occur spontaneously but it could also be catalyzed by glutamine cyclotransferase as proposed¹¹, and N-terminal methylation could be catalyzed by an uncharacterized plant peptide N-methyltransferase. In general, the biosynthesis of BURP-domain-derived plant RiPPs differs from the biosynthesis of head-to-tail-cyclic plant RiPPs such as cyclotides in that the major macrocyclization step for BURP-domain peptides is the first biosynthetic step, whereas it is the last step for cyclotide biosynthesis^{3–9}. BURP-domain cyclases are the second example of autocatalytic enzymes involved in RiPP biosynthesis after fungal peptide N-methyltransferases⁴⁹. The structures of BURP-domain-derived plant peptides are similar in macrocyclization chemistry to radical SAM peptide cyclase products^{16,17,22}. As all BURP-domain cyclizations in this study were accomplished under aerobic conditions, BURP domains seem to catalyze their crosslinks in the presence of oxygen, whereas radical SAM peptide cyclases require anaerobic reaction conditions due to their Fe-S-cluster⁵⁰. Mechanistic studies on BURP-domain enzymes will shed light on BURP-domain catalysis and enable a better comparison to radical SAM enzymes regarding peptide macrocyclization. Although we only present single-turnover catalysis by BURP domains in this study, the multi-core-peptide motifs of BURP domains indicate that these proteins can catalyze multiple turnovers.

With an average of nine BURP-domain loci in each analyzed plant genome, BURP domains are a common protein family within plant genomes and a significant fraction of BURP domains seems to be involved in peptide biosynthesis. Gene expression studies of BURP-domain precursor peptides such as AhyBURP-homolog Sali3-2 and GLYMA_04G180400 point to abiotic stress-related functions, as these peptide precursors are upregulated during increased soil acidity and salinity^{43,51} in soybeans. BURP-domain peptide products could have potential roles in alleviating such stresses, for example by metal chelation. The association of an insecticidal cyclopeptide alkaloid to a gene encoding a BURP-domain precursor peptide in mung bean also suggests that side-chain-macrocyclic plant peptides are involved in biotic plant defense³⁸. Given the connection of BURP domains and peptide biosynthesis, the endogenous

functions of BURP-domain-derived peptides in plant physiology and development should be the focus of future studies. With growing plant genomic resources, we hope this work inspires increased discovery of side-chain-macrocyclic peptides and underlying BURP-domain cyclases for applications in medicine and agriculture.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41589-021-00892-6>.

Received: 17 April 2021; Accepted: 2 September 2021;

Published online: 22 November 2021

References

1. Tan, N. H. & Zhou, J. Plant cyclopeptides. *Chem. Rev.* **106**, 840–895 (2006).
2. de Veer, S. J., Kan, M.-W. & Craik, D. J. Cyclotides: from structure to function. *Chem. Rev.* **119**, 12375–12421 (2019).
3. Arnison, P. G. et al. Ribosomally synthesized and post-translationally modified peptide natural products: overview and recommendations for a universal nomenclature. *Nat. Prod. Rep.* **30**, 108–160 (2013).
4. Jennings, C., West, J., Waine, C., Craik, D. & Anderson, M. Biosynthesis and insecticidal properties of plant cyclotides: the cyclic knotted proteins from *Oldenlandia affinis*. *Proc. Natl Acad. Sci. USA* **98**, 10614–10619 (2001).
5. Mylne, J. S. et al. Albumins and their processing machinery are hijacked for cyclic peptides in sunflower. *Nat. Chem. Biol.* **7**, 257–259 (2011).
6. Nguyen, G. K. T. et al. Butelase 1 is an Asx-specific ligase enabling peptide macrocyclization and synthesis. *Nat. Chem. Biol.* **10**, 732–738 (2014).
7. Harris, K. S. et al. Efficient backbone cyclization of linear peptides by a recombinant asparaginyl endopeptidase. *Nat. Commun.* **6**, 10199 (2015).
8. Chekan, J. R., Estrada, P., Covello, P. S. & Nair, S. K. Characterization of the macrocyclase involved in the biosynthesis of RiPP cyclic peptides in plants. *Proc. Natl Acad. Sci. USA* **114**, 6551–6556 (2017).
9. Barber, C. J. S. et al. The two-step biosynthesis of cyclic peptides from linear precursors in a member of the plant family Caryophyllaceae involves cyclization by a serine protease-like enzyme. *J. Biol. Chem.* **288**, 12500–12510 (2013).
10. Hattori, J., Boutillier, K. A., Campagne, M. M. L. & Miki, B. L. A conserved BURP domain defines a novel group of plant proteins with unusual primary structures. *Mol. Gen. Genet. MGG* **259**, 424–428 (1998).
11. Kersten, R. D. & Weng, J. K. Gene-guided discovery and engineering of branched cyclic peptides in plants. *Proc. Natl Acad. Sci. USA* **115**, E10961–E10969 (2018).
12. Treacy, B. K. et al. Bnm1, a brassica pollen-specific gene. *Plant Mol. Biol.* **34**, 603–611 (1997).
13. Bassünér, R. et al. Abundant embryonic mRNA in field bean (*Vicia faba* L.) codes for a new class of seed proteins: cDNA cloning and characterization of the primary translation product. *Plant Mol. Biol.* **11**, 321–334 (1988).
14. Yamaguchi-Shinozaki, K. & Shinozaki, K. The plant hormone abscisic acid mediates the drought-induced expression but not the seed-specific expression of Rd22, a gene responsive to dehydration stress in *Arabidopsis thaliana*. *Mol. Gen. Genet. MGG* **238**, 17–25 (1993).
15. Zheng, L., Heupel, R. C. & DellaPenna, D. The beta subunit of tomato fruit polygalacturonase isoenzyme 1: isolation, characterization, and identification of unique structural features. *Plant Cell* **4**, 1147–1156 (1992).
16. Schramma, K. R., Bushin, L. B. & Seyedsayamdost, M. R. Structure and biosynthesis of a macrocyclic peptide containing an unprecedented lysine-to-tryptophan crosslink. *Nat. Chem.* **7**, 431–437 (2015).
17. Imai, Y. et al. A new antibiotic selectively kills Gram-negative pathogens. *Nature* **576**, 459–464 (2019).
18. Nagano, N. et al. Class of cyclic ribosomal peptide synthetic genes in filamentous fungi. *Fungal Genet. Biol.* **86**, 58–70 (2016).
19. Koiso, Y. et al. Ustiloxin: a phytotoxin and a mycotoxin from false smut balls on rice panicles. *Tetrahedron Lett.* **33**, 4157–4160 (1992).
20. Culvenor, C. C. J. et al. Structure elucidation and absolute configuration of phomopsis A, a hexapeptide mycotoxin produced by phomopsis leptostromiformis. *Tetrahedron* **45**, 2351–2372 (1989).
21. Ding, W. et al. Biosynthetic investigation of phomopsins reveals a widespread pathway for ribosomal natural products in Ascomycetes. *Proc. Natl Acad. Sci. USA* **113**, 3521–3526 (2016).
22. Nguyen, T. Q. N. et al. Post-translational formation of strained cyclophanes in bacteria. *Nat. Chem.* <https://doi.org/10.1038/s41557-020-0519-z> (2020).
23. Nicholas, A. et al. Total synthesis and stereochemical assignment of streptide. *J. Am. Chem. Soc.* **141**, 17361–17369 (2019).

24. Shabani, S., White, J. M. & Hutton, C. A. Total synthesis of the putative structure of asperipin-2a and stereochemical reassignment. *Organ. Lett.* **22**, 7730–7734 (2020).
25. Ye, Y. et al. Heterologous production of asperipin-2a: proposal for sequential oxidative macrocyclization by a fungi-specific DUF3328 oxidase. *Org. Biomol. Chem.* **17**, 39–43 (2019).
26. Crüsemann, M. et al. Heterologous expression, biosynthetic studies, and ecological function of the selective Gq-signaling inhibitor FR900359. *Angew. Chemie Int. Ed.* **57**, 836–840 (2018).
27. Horai, H. et al. MassBank: a public repository for sharing mass spectral data for life sciences. *J. Mass Spectrom.* **45**, 703–714 (2010).
28. Smith, C. A. et al. METLIN: a metabolite mass spectral database. *Ther. Drug Monit.* **27**, 747–751 (2005).
29. Wang, M. et al. Sharing and community curation of mass spectrometry data with Global Natural Products Social Molecular Networking. *Nat. Biotechnol.* **34**, 828–837 (2016).
30. Kind, T. et al. Identification of small molecules using accurate mass MS/MS search. *Mass Spectrom. Rev.* **37**, 513–532 (2018).
31. Behsaz, B. et al. De novo peptide sequencing reveals many cyclopeptides in the human gut and other environments. *Cell Syst.* **10**, 99–108.e5 (2020).
32. Ng, J. et al. Dereplication and de novo sequencing of nonribosomal peptides. *Nat. Methods* **6**, 596–599 (2009).
33. Dührkop, K. et al. SIRIUS 4: a rapid tool for turning tandem mass spectra into metabolite structure information. *Nat. Methods* **16**, 299–302 (2019).
34. Dührkop, K. et al. Systematic classification of unknown metabolites using high-resolution fragmentation mass spectra. *Nat. Biotechnol.* **9**, 462–471 (2020).
35. Falick, A. M., Hines, W. M., Medzihradsky, K. F., Baldwin, M. A. & Gibson, B. W. Low-mass ions produced from peptides by high-energy collision-induced dissociation in tandem mass spectrometry. *J. Am. Soc. Mass. Spectrom.* **4**, 882–893 (1993).
36. Phillipson, J. D. Chemical investigations of herbarium material for alkaloids. *Phytochemistry* **21**, 2441–2456 (1982).
37. Nguyen, G. K. T. et al. Discovery and characterization of novel cyclotides originated from chimeric precursors consisting of albumin-1 chain a and cyclotide domains in the fabaceae family. *J. Biol. Chem.* **286**, 24275–24287 (2011).
38. Sugawara, F. et al. Insecticidal peptide from mungbean: a resistant factor against infestation with azuki bean weevil. *J. Agric. Food Chem.* **44**, 3360–3364 (1996).
39. Griesmann, M. et al. Phylogenomics reveals multiple losses of nitrogen-fixing root nodule symbiosis. *Science* **361**, eaat1743 (2018).
40. Zhuang, W. et al. The genome of cultivated peanut provides insight into legume karyotypes, polyploid evolution and crop domestication. *Nat. Genet.* **51**, 865–876 (2019).
41. Sainsbury, F., Thuenemann, E. C. & Lomonossoff, G. P. pEAQ: versatile expression vectors for easy and quick transient expression of heterologous proteins in plants. *Plant Biotechnol. J.* **7**, 682–693 (2009).
42. Boersema, P. J., Raijmakers, R., Lemeer, S., Mohammed, S. & Heck, A. J. R. Multiplex peptide stable isotope dimethyl labeling for quantitative proteomics. *Nat. Protoc.* **4**, 484–494 (2009).
43. Tang, Y. et al. Expression of a vacuole-localized BURP-domain protein from soybean (SAL13-2) enhances tolerance to cadmium and copper stresses. *PLoS ONE* **9**, e107337 (2014).
44. Ragland, M. & Soliman, K. M. A molecular approach to understanding aluminum tolerance in soybean (*Glycine max* L.). *Global Environmental Biotechnology* (ed. Wise, D. L.) Vol. 114, 125–138 (Springer, 1997).
45. Goodstein, D. M. et al. Phytozome: a comparative platform for green plant genomics. *Nucleic Acids Res.* **40**, D1178–D1186 (2012).
46. Kautsar, S. A., Suarez Duran, H. G., Blin, K., Osbourn, A. & Medema, M. H. PlantSMASH: automated identification, annotation and expression analysis of plant biosynthetic gene clusters. *Nucleic Acids Res.* **45**, W55–W63 (2017).
47. Schmutz, J. et al. Genome sequence of the palaeopolyploid soybean. *Nature* **463**, 178–183 (2010).
48. Medema, M. H. & Osbourn, A. Computational genomic identification and functional reconstitution of plant natural product biosynthetic pathways. *Nat. Prod. Rep.* **33**, 951–962 (2016).
49. van der Velden, N. S. et al. Autocatalytic backbone N-methylation in a family of ribosomal peptide natural products. *Nat. Chem. Biol.* **13**, 833–835 (2017).
50. Broderick, J. B., Duffus, B. R., Duschene, K. S. & Shepard, E. M. Radical S-adenosylmethionine enzymes. *Chem. Rev.* **114**, 4229–4317 (2014).
51. Xu, H. et al. Genome-scale identification of Soybean BURP domain-containing genes and their expression under stress treatments. *BMC Plant Biol.* **10**, 197 (2010).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2021

Methods

Materials. All chemicals were purchased from Fisher Scientific, unless otherwise noted. Oligonucleotide primers were purchased from Integrated DNA Technologies, Inc. Synthetic genes were purchased as gBlocks from Integrated DNA Technologies, Inc. or as GenParts or synthetic genes from GenScript, Inc. Solvents for LC–MS were Optima LC–MS grade (Fisher Scientific). We carried out high-resolution MS analysis using a Thermo ESI-Q-Exactive Orbitrap mass spectrometer coupled to a Thermo Vanquish ultra-HPLC system. NMR analysis was conducted on a Bruker Ascend 800 MHz NMR spectrometer equipped with a 5 mm Triple Resonance Inverse Detection TCI CryoProbe. Preparative and semipreparative HPLC was performed on a Shimadzu LC-10AP liquid chromatograph equipped with a SPD-20A ultraviolet-visible light detector and a FRC-10A fraction collector.

Plant material. Botanical garden plant samples were collected from the Matthaei Botanical Garden at the University of Michigan, Ann Arbor, in August 2019 and August 2020. Herbarium plant samples were collected from the Herbarium of the University of Michigan, Ann Arbor (Supplementary Dataset 2). A *S. kraussiana* var. *aurea* plant for RNA isolation was purchased from the Matthaei Botanical Garden, Ann Arbor. *S. kraussiana* var. *aurea* plants for chemical isolation were purchased from Plants4Home LLC. *M. sativa* (alfalfa) seeds for chemical extraction and *Glycine max* seeds for cultivation were purchased from Outsidepride. *A. hypogaea* seeds for cultivation were purchased from the KAFA company. *Amaranthus hypochondriacus* and *Celosia argentea* seeds were purchased from Seedville. *Lycium barbarum* root was purchased from 9GreenBox. A *Stephanotis floribunda* plant was purchased from Emerald Goddess Gardens. *C. americanus* root was purchased from Starwest Botanicals. *Rubia cordifolia* root powder was purchased from MB Herbs. *N. benthamiana* seeds for cultivation were a gift from the Weng laboratory (Whitehead Institute, MIT). *E. altensteinii* leaves were a gift from the Matthaei Botanical Garden, Ann Arbor. *C. canadensis* leaf and stem tissue for RNA isolation and chemical purification were collected at 428 Church Street, Ann Arbor, MI 48109, USA.

Plant growth. *N. benthamiana*, *A. hypogaea* and *G. max* were grown in SunGro Sunshine Mix 4 soil with added vermiculite (Plantain Inc.) and added Osmocote fertilizer under plant growth lights with a 16 h light/8 h dark cycle for 3 months.

Peptide chemotyping. For botanical garden samples, 0.2 g of plant material (fresh weight) was collected from aerial tissues, frozen and ground with a MP Biomedicals FastPrep-24-5G Tissuelyser with a Coolprep cryogenic adapter in 2-ml MP Biomedicals tubes with 2.3-mm Zirconia beads (Fisher Scientific). For herbarium specimen, 0.1 g of plant material (dry weight) was collected from aerial tissues (stem and leaf, if available) and ground as described above. For reference peptide chemotyping, 0.2 g of *A. hypochondriacus* seeds, *C. argentea* seeds, *S. floribunda* leaves, *R. cordifolia* root powder, *C. americanus* roots and *L. barbarum* roots were sampled in triplicate and ground as described above. For chemotyping of desmethyl-selanines, 1 g of *S. kraussiana* leaves were sampled. Ground plant material was extracted with 3 ml of methanol for 1 h at 37 °C in a glass vial (Fisher Scientific 03-337-26) and crude methanol extracts were dried under nitrogen gas in a separate glass vial. Dried plant methanol extracts were resuspended in water (3 ml) and partitioned twice with hexane (3 ml), twice with ethyl acetate (3 ml) and once with *n*-butanol (3 ml). For the botanical garden and herbarium samples, the *n*-butanol fraction was dried by vacuum centrifugation in a Thermo Scientific SPD140P1 speedvac and resuspended in 2 ml of methanol for LC–MS analysis. For reference peptide chemotyping, hexane fractions and ethyl acetate fractions were dried under nitrogen and resuspended in 2 ml of methanol, and *n*-butanol fractions were dried in vacuo as described above and resuspended in 2 ml of methanol. All peptide extracts were filtered by Whatman syringeless filters (0.2 µm) and were subjected to LC–MS/MS analysis with the following parameters: injection volume 5 µl; LC, Phenomenex Kinetex 2.6 µm C18 reverse phase 100 Å 150 × 3 mm LC column; LC gradient, solvent A, 0.1% formic acid; solvent B, acetonitrile (0.1% formic acid); 0 min, 10% B; 5 min, 60% B; 5.1 min, 95% B; 6 min, 95% B; 6.1 min, 10% B; 9.9 min, 10% B; 0.5 ml min⁻¹; MS, positive ion mode; full MS, resolution 70,000; mass range 400–1,200 *m/z*; dd-MS² (data-dependent MS/MS), resolution 17,500; AGC target 1 × 10⁵, loop count 5, isolation width 1.0 *m/z*, collision energy 25 eV and dynamic exclusion 0.5 s. Reference peptide samples were also analyzed with collision energies of 10, 20, 30 or 40 eV. LC–MS data were analyzed with QualBrowser in the Thermo Xcalibur software package (v.4.3.73.11, Thermo Scientific). For characterization of desmethyl-selanines in *S. kraussiana* extracts, MS settings were as follows: TargetSIM-DDA, MS¹, isolation width 4 *m/z* with target masses 539.2864 (desmethyl-selanine B) and 884.4189 (desmethyl-selanine A); dd-MS², HCD 25 eV; MS², isolation width 0.4 *m/z* and AGC target, 1 × 10⁵. Reference peptide quantification was done by peak area integration from extracted ion chromatograms (EIC) from three samples of biologically different source plant or different batches of plant material. Statistical analysis of EIC peak areas was done with GraphPrism 9.

MS data analysis. Peptide chemotyping RAW files from the botanical garden and herbarium samples were filtered for MS² spectra with unmethylated,

monomethylated and dimethylated iminium ions (Supplementary Table 1) among the 20 most intense fragment ion mass peaks with an allowed mass error of 10 ppm and converted to centroided mzXML-files by the command line version of Msconvert⁵² using the following command:

```
msconvert.exe *.RAW --filter "peakPicking true [1,2]" --filter "mzPresent [58.06513,60.04439,70.06513,72.08078,74.06004,84.04439,84.08078,86.09643,87.05529,88.0393,88.07569,100.11208,101.07094,101.10732,102.05495,102.09134,104.05285,110.07127,114.12773,115.08659,115.12297,116.0706,118.0685,120.08078,124.08692,129.10224,129.11347,129.13862,130.08625,132.08415,134.09643,136.07569,138.10257,143.12912,148.11208,150.09134,157.14477,159.09167,164.10699,173.10732,187.12297]" mzTol=10 ppm threshold=20" --filter "msLevel 2" --mzXML -o (output directory)
```

A molecular network was created for the Matthaei Botanical Garden samples and the herbarium samples using the online workflow (<https://ccms-ucsd.github.io/GNPSDocumentation/>, release 26)³⁹ with Msconvert-processed reference plant peptide MS² spectra (Supplementary Table 1) in a separate networking group from the datasets of the botanical garden or the herbarium on the GNPS website (<http://gnps.ucsd.edu>). The data were filtered by removing all MS/MS fragment ions within ±17 Da of the precursor *m/z*. MS/MS spectra were window filtered by choosing only the top six fragment ions in the ±50 Da window throughout the spectrum. The precursor ion mass tolerance was set to 0.05 Da and a MS/MS fragment ion tolerance of 0.25 Da. A network was then created where edges were filtered to have a cosine score above 0.7 and more than five matched peaks. Further, edges between two nodes were kept in the network if and only if each of the nodes appeared in each other's respective top ten most similar nodes. Finally, the maximum size of a molecular family was set to 500, and the lowest scoring edges were removed from molecular families until the molecular family size was below this threshold. The spectra in the network were then searched against GNPS spectral libraries. The library spectra were filtered in the same manner as the input data. All matches kept between network spectra and library spectra were required to have a score above 0.7 and at least five matched peaks. Molecular networking data were analyzed with Cytoscape (v.3.8.0)³³. Candidate peptide clusters were identified by the presence of reference plant peptide spectra or by manual analysis for iminium ions. Unfiltered metabolomic datasets were also analyzed for candidate peptide analytes manually by searching MS² data in Qualbrowser via the 'ms2' scan filter for spectra with iminium ions of tyrosine (136.0757 Da) and tryptophan (159.0917 Da). Candidate peptide spectra derived from molecular networking and manual iminium ion filtering were analyzed for all iminium ions from Supplementary Table 1 in Qualbrowser.

Transcriptomics. *S. kraussiana* aerial total RNA was extracted from aerial tissue with the QIAGEN RNeasy Plant Mini kit. RNA quality was assessed by Agilent Bioanalyzer. Strand-specific messenger RNA libraries were prepared with TruSeq Stranded Total RNA with Ribo Zero Library Preparation Kit (Illumina) and sequenced with a HiSeq2500 Illumina sequencer (150 × 150 bp). Illumina sequence fastq-files were assembled by SPAdes (v.3.14.0)⁵⁴. BURP-domain transcripts were searched in the de novo transcriptome by querying Sali3-2 (ref. ⁴³) (GenBank NP_001235780.2) by tblastn algorithm⁵⁵ on an internal BLAST server⁵⁶. To clone and sequence candidate peptide precursor gene *SkrBURP*, complementary DNA was prepared from *S. kraussiana* aerial total RNA with SuperScript III First-Strand Synthesis System (Invitrogen). Candidate *SkrBURP* transcripts were used to design cloning primers (*SkrBURP*-pEAQ-HT-fwd: TGCCCAAATTCGCGACCGGTATGGCTCAATCACTATTCTATTG, *SkrBURP*-pEAQ-HT-rev: CCAGAGTTAAAGGCCTCGAGTCAAGAAGAAC ATCGCGCGTC) for amplification of *SkrBURP* with Phusion High-Fidelity DNA polymerase (New England Biolabs). *SkrBURP* was cloned into pEAQ-HT⁴¹, which was linearized by restriction enzymes AgeI and XhoI, by Gibson cloning assembly (New England Biolabs)³⁷. Cloned *SkrBURP* was sequenced by Sanger sequencing.

Genomics. For *C. canadensis* genome mining, BURP-domain transcripts were searched in *C. canadensis* genome (GenBank [GCA_003255065.1](https://www.ncbi.nlm.nih.gov/nuclot/003255065.1)) by querying Sali3-2 (ref. ⁴³) using tblastn algorithm⁵⁵. Candidate BURP-domain genes were translated by European Molecular Biology Laboratory (EMBL) Transeq⁵⁸ into protein sequences and analyzed for candidate core peptide sequences matching observed amino acid iminium ions and parent masses in MS/MS and MS spectra, respectively. *C. canadensis* young leaf RNA was extracted from aerial tissue with the QIAGEN RNeasy Plant Mini kit and corresponding cDNA was prepared with SuperScript III First-Strand Synthesis System (Invitrogen). Candidate precursor-peptide genes CcaBURP1 and CcaBURP2 were used to design cloning primers (CcaBURP1-pEAQ-HT-fwd: TGCCCAAATTCGCGACCGGTATGAAGCTTCATCGCCTAAC, CcaBURP1-pEAQ-HT-rev: CCAGAGTTAAAGGCCTCGAGTTAAGTAG TTTTCTGGGGAGTCC, CcaBURP2-pEAQ-HT-fwd: TGCCCAAATTCGCGACCGGTATGAAGCCTCATCGCCTAC, CcaBURP2-pEAQ-HT-rev: CCAGAGTTAAAGGCCTCGAGTTAAGTAGTCTTATGGGGAATCC) for amplification of CcaBURP1 and CcaBURP2, respectively, with Phusion High-Fidelity DNA polymerase (New England Biolabs). CcaBURP1 and

CcaBURP2 were cloned into pEAQ-HT⁴¹, which was linearized by restriction enzymes AgeI and XhoI, by Gibson cloning assembly (New England Biolabs)³⁷. Cloned CcaBURP1 and CcaBURP2 were sequenced by Sanger sequencing. *A. hypogaea* stem RNA was extracted with the QIAGEN RNeasy Plant Mini kit and corresponding cDNA was prepared with SuperScript III First-Strand Synthesis System (Invitrogen). For *A. hypogaea* genome mining, BURP-domain genes were searched in *A. hypogaea* genome (JGI Phytozome⁴⁵; *A. hypogaea* v.1.0) by keyword search 'BURP'. Candidate BURP-domain genes were translated by EMBL Transeq into protein sequences and analyzed for candidate core peptide sequences matching observed amino acid iminium ions and parent masses in MS/MS and MS spectra, respectively. Candidate precursor-peptide gene *AhyBURP* (*arahy.Tifrunner.gnm1.ann1.316GZV*) was used to design cloning primers (AhyBURP-pEAQ-HT-fwd: TGCCCAAAATTCGCGACCGGTATGGAGTTCGAGTGTGCTGTGTC, AhyBURP-pEAQ-HT-rev: CCAGAGTTAAAGGCCTCGAGTTAACTT GCCATATTGGCATTATTG) for amplification of *AhyBURP* with Phusion High-Fidelity DNA polymerase (New England Biolabs). *AhyBURP* was cloned into pEAQ-HT⁴¹, which was linearized by restriction enzymes AgeI and XhoI, by Gibson cloning assembly (New England Biolabs)³⁷.

Peptide classification, isolation and structure elucidation. Candidate peptides were connected to a matching candidate core peptide sequence in BURP-domain proteins based on the observed iminium ions, observed parent mass and potential posttranslational modifications of a candidate core peptide. Herein, modifications to yield common structural features of plant peptides (Supplementary Dataset 1) were tested: cyclization (loss of two hydrogens per cyclization), N-terminal mono- or dimethylation (addition of 1 or 2 methyl groups), N-terminal pyroglutamate formation from N-terminal glutamine (loss of ammonia), C-terminal decarboxylation and dehydrogenation (loss of formic acid). Core peptide sequences were searched in the N-terminal half of BURP-domain sequences based on known BURP-domain peptide precursors⁴¹. If a pyroglutamate iminium ion was detected, core peptide sequences with an N-terminal glutamine were tested. C-termini of candidate core peptides were preferentially tyrosine or tryptophan. Candidate cyclic peptide analytes matching to a core peptide in a BURP domain were classified, if applicable, by the presence of reference peptide spectra (Supplementary Table 11) in their respective molecular cluster, purified and structurally elucidated as specified in Supplementary Note 2. NMR data analysis was done with Bruker Topspin (v.4.0.8). Marfey's analysis and chiral HPLC of hydrolyzed amino acids was performed as follows: 0.3 mg of a target peptide were dissolved in 1 ml of 6 N hydrochloric acid (HCl) and stirred at 110 °C for 4 h. The hydrolysate was dried in vacuo and resuspended in 2 ml of water and dried in vacuo twice. The dried hydrolysate was dissolved in 2 ml of water and applied to a Strata C8 SPE column (Phenomenex, 100 mg) and eluted with 50% acetonitrile. The eluate was dried in vacuo, resuspended in 100 µl of water and it was added with 200 µl of Marfey's reagent (Thermo Scientific, 1% (w/v) acetone), added with 40 µl of sodium bicarbonate (1 M), and incubated at 40 °C for 1 h. The reaction was quenched with 20 µl of 2 N HCl, diluted 1:1 with 50% acetonitrile and subjected to LC-MS analysis as for peptide chemotyping with the mass range: 100–800 *m/z*. Geometry optimization of macrocyclization stereoisomers of cercic acid and leguminin for comparison to nuclear Overhauser enhancement spectroscopy NMR data was performed as follows: stereoisomer structures were energy-minimized by MM2 and subsequent MMFF94 calculation in Chem3D (v.18.1, maximum iterations 10,000). The resulting conformers were geometry-optimized by B3LYP-D3/6-31+G(d,p) method in Gaussian 16 (revision A.03)³⁹ on the Great Lakes Advanced Research Computing cluster at the University of Michigan and analyzed in Chem3D.

Antibacterial assays. Each tested bacterium (*Bacillus subtilis* 168, *Staphylococcus aureus* ATCC43300, *Shigella flexneri* BS103) was streaked onto a tryptic soy agar plate and was incubated at 37 °C for 24 h. One colony was then transferred to 3 ml of fresh Luria-Bertani medium (for *B. subtilis*), brain heart infusion broth (for *S. aureus*) or 2xTY broth (for *S. flexneri*), incubated at 37 °C for 24 h and the cell density was adjusted to 10⁴–10⁵ colony forming units per ml using the respective broth media. The compounds to be tested were dissolved in dimethylsulfoxide (DMSO) and diluted with water to give 2 mM stock solution (20% DMSO). An aliquot (10 µl) of each compound was transferred to a 96-well microtiter plate, and freshly prepared microbial broth (190 µl) was added to each well to give final compound concentrations of 100 µM in 1% DMSO. The plates were incubated at 37 °C for 24 h, and the optical density of each well was measured spectrophotometrically at 600 nm (OD_{600nm}) using M5 spectramax plate reader with SoftMax Pro Software (Molecular Devices LLC). Rifampicin and ciprofloxacin were used as positive controls (100 and 60 µM, respectively). Meanwhile, broth medium with and without bacterial inoculation were tested as negative controls. Data were processed using GraphPad Prism v.9.2.2 (GraphPad Software Inc.) (Extended Data Fig. 6).

Anticancer assays. *Cancer cell lines.* LNCaP, Caco-2, Huh7 and H1437 cells were maintained at 37 °C and 5% CO₂ atmosphere. LNCaP and H1437 were grown in Roswell Park Memorial Institute 1640 Medium (Fisher Scientific, 11-875-119), Huh7 and Caco-2 were grown in Dulbecco's Modified Eagle Medium (Thermo

Fisher, 11995073). Both lots of base media were supplemented with 10% FBS (Fisher Scientific, 10-437-028) and 1× penicillin-streptomycin (Gibco, 15140122).

Cancer inhibition bioassay. 384-well plates (Perkin Elmer, 6057300) were seeded with LNCaP (8,000 cells per well), Caco-2 (5,000 cells per well), Huh7 (3,000 cells per well) or H1437 (7,000 cells per well) in 50 µl of their respective complete media. These seeding densities were optimized such that the cells would reach 90–100% confluency after 72 h. Cells were allowed to attach overnight at 37 °C and 5% CO₂. Following attachment, compounds (10 mg ml⁻¹) were added using an HPD300e digital compound dispenser in ten-point, twofold dose response (*N* = 3) from a top concentration of 100 µg ml⁻¹. Compounds were block randomized and normalized to a final DMSO concentration of 1%. After compound addition, cells were incubated at 37 °C and 5% CO₂ for 48 h. Each cell line had *N* = 68 vehicle (DMSO) control wells. Cells were fixed with 4% paraformaldehyde for 30 min, permeabilized with 0.03% Triton X-100 for 20 min and rinsed twice with PBS. After washes, cells were stained with a 1:1,000 PBS dilution of Hoechst 33340 trihydrochloride, trihydrate (Thermo Scientific, H1399) for 30 min to label cell nuclei. Cells were then washed twice with PBS and left in a final volume of 50 µl per well. Fixed and stained plates were imaged at ×10 magnification using a Thermo Fisher CX5 high content imaging microscope with LED excitation (386/23 nm). Exposure time was optimized to maximize signal/background for each cell line. Nine fields per well were collected at a single *z*-plane as determined by the image-based autofocus.

Bioassay analysis. Cell viability was determined via image-based analysis and used to detect compound induced cell inhibition. Cell nuclei were segmented and counted using the open source image analysis software CellProfiler. A percentage viability score for compounds in each cell line was generated by normalizing all of the data to the average of the well-level cell counts for the vehicle control (DMSO). Unpaired two-tailed *t*-tests were used to statistically compare the compounds to the vehicle control and were performed in GraphPad Prism 9 (Graphpad Software).

Transient gene expression in *N. benthamiana*. Wild-type *SkrBURP*, *CcaBURP1*, *CcaBURP2*, *AhyBURP* and synthetic genes *SkrBURP*-[1xVLFYPSY], *CcaBURP1*-[1xQILFW], *CcaBURP2*-[1xQLLVW] (Supplementary Note 11) were each transiently expressed as pEAQ-HT-constructs in *N. benthamiana* leaves (4–6-week-old plants) via *Agrobacterium tumefaciens* LBA4404 syringe infiltration⁴¹ as previously described⁴¹. Six days after infiltration, 0.2 g of *N. benthamiana* leaves (fresh weight) were collected, placed into a 2-ml Tissuelyzer tube filled with ten zirconia/silica beads (2.3 mm in diameter) and frozen on dry ice. Frozen *N. benthamiana* samples were then lysed with a Fastprep-24 5G Tissuelyser (MP Biomedicals) by three 40 s cycles at 6 m s⁻¹ speed. Then 1 ml of 80% methanol was subsequently added to the tubes with the ground *N. benthamiana* tissue, lysed for one additional cycle on the Tissuelyser and incubated for 10 min at 60 °C. Crude *N. benthamiana* extracts were centrifuged at 16,000g for 5 min, filtered by Whatman syringeless filters (0.2 µm) and were subjected to LC-MS/MS analysis as specified for peptide chemotyping. For characterization of desmethyl-selanines via reductive dimethylation, *SkrBURP* was transiently expressed in roughly 50 g of *N. benthamiana* leaves for 7 d, which were subsequently harvested, ground and extracted with methanol for 24 h at 37 °C. Crude methanol extracts were filtered, dried in vacuo, resuspended in 100 ml of water, partitioned twice with 100 ml of hexane, partitioned twice with 100 ml of ethyl acetate and extracted once with 100 ml of *n*-butanol. The *n*-butanol extract was dried in vacuo, resuspended in 5 ml of 10% acetonitrile (0.1% trifluoroacetic acid (TFA)), and subjected to preparative HPLC (see the in vitro scaled enzyme assays for HPLC settings). HPLC fractions were analyzed for desmethyl-selanine A and B by LC-MS and corresponding fractions were dried in vacuo. Reductive dimethylation was performed as described⁴². Briefly, desmethyl-selanine-containing HPLC fractions were redissolved in 100 µl of 100 mM sodium acetate buffer (pH 6, 100 µl volume). Then 51 µl of peptide solution were added with 3 µl of 4% formaldehyde solution and 3 µl of sodium cyanoborohydride solution (260 mM). After 5 min of incubation at 37 °C, 3 µl of ammonium hydroxide (4%) were added and the reaction mixture was analyzed by LC-MS as specified for peptide chemotyping.

Protein expression in *E. coli*. *AhyBURP* was synthesized as an *E. coli*-codon-optimized gene, cloned into pHis8 (pET28a with a 8×His-tag), linearized with NcoI, via Gibson cloning³⁷. *CcaBURP1*-[1xQILFW], *CcaBURP2*-[1xQLLVW], *SkrBURP*-[1xVLFYPSY] were synthesized as *E. coli*-codon-optimized genes (Supplementary Note 11), cloned into pvp57k (pvp56k-based vector without BarCat cassette³⁸), linearized with SpeI, via Gibson cloning³⁷. Corresponding expression vectors were transformed into *E. coli* BL21 (DE3) by heat shock. Two 10-ml starter cultures each containing 10 ml of sterile liquid Luria-Bertani medium with 50 µg ml⁻¹ kanamycin sulfate were inoculated for each *E. coli* BL21 (DE3) expression strain from transformant colonies and incubated for 16 h at 37 °C and 220 r.p.m. Two 1-l batches of sterile liquid terrific broth medium with 50 µg ml⁻¹ kanamycin sulfate in 2.8-l Fernbach flasks were inoculated with the starter cultures and incubated at 37 °C and 200 r.p.m. until culture OD_{600nm} reached 1. The culture was then induced with 50 µM IPTG and incubated at 18 °C and 200 r.p.m. for

16 h. The culture was centrifuged at 3,800g and 4°C for 30 min. Supernatant was discarded and cell pellets were resuspended in 5 ml of lysis buffer (50 mM Tris-HCl, pH 8, 500 mM sodium chloride, 20 mM imidazole, 0.1% Tween-20, 10 mM β -mercaptoethanol (Sigma), 10% glycerol) per gram of wet cell pellet. Lysozyme was added at a ratio of 1.5 mg ml⁻¹, about 10 U of DNase I was added to the cell suspension and it was stirred at 4°C for 1 h. The cell suspension was sonicated on ice for 8 min at 80% amplitude (2 s on, 2 s off). The resulting lysates were centrifuged at 19,000g and 4°C for 45 min.

For AhyBURP purification, the supernatant lysate was separated from the cell pellet with inclusion bodies. Inclusion bodies were resuspended in denaturing lysis buffer (50 mM Tris-HCl (pH 8), 500 mM sodium chloride, 10 mM imidazole, 10% glycerol, 0.5% Tween-20, 0.1% Triton X-100, 10 mM β -mercaptoethanol (Sigma-Aldrich), 8 M urea). The inclusion body resuspension was stirred at room temperature for 1–2 h, while the clarified lysate was adjusted to 8 M urea and also stirred at room temperature. The denaturing inclusion body resuspension was centrifuged at 20,000g and 20°C for 30 min. The clarified lysate from inclusion bodies was combined with the denatured supernatant lysate. Then 2 ml of equilibrated Ni-NTA were used per liter of culture, and the denatured clarified lysate was poured over Ni-NTA in 20-ml disposable plastic columns (Bio-Rad) at room temperature. Flowthrough was discarded, and each 2 ml of resin was washed with 25 ml of wash buffer (50 mM Tris-HCl (pH 8), 500 mM sodium chloride, 20 mM imidazole, 10% glycerol, 10 mM β -mercaptoethanol, 8 M urea). AhyBURP was eluted from Ni-NTA with 20 ml of elution buffer per 2 ml of resin (50 mM Tris-HCl (pH 8), 500 mM sodium chloride, 250 mM imidazole, 10% glycerol, 10 mM β -mercaptoethanol, 8 M urea). Eluted protein was immediately placed in dialysis tubing (3.5 kDa molecular weight cutoff (MWCO)) and dialyzed in dialysis buffer (50 mM Tris-HCl (pH 8), 250 mM sodium chloride, 10 mM β -mercaptoethanol, 10% glycerol, 1 M urea) overnight at 4°C. The next day, the protein was dialyzed in the same conditions at 4°C, but without urea. After 5 h, protein was dialyzed in a third and final dialysis buffer with tobacco etch virus (TEV) protease (50 mM Tris-HCl, 100 mM sodium chloride, 2% glycerol, 3 mM reduced glutathione and 0.3 mM oxidized glutathione). TEV protease was added to AhyBURP in a ratio of 1 mg of TEV protease per 40 mg of AhyBURP. The protein and TEV protease dialyzed overnight at 4°C. To remove precipitate the next day, the protein sample was centrifuged at 20,000g for 40 min at 4°C. The sample was poured over equilibrated Ni-NTA and flowthrough was collected and concentrated in 10-kDa MWCO concentrators to yield concentrated AhyBURP for enzyme assays or storage at -80°C. For scaled AhyBURP expression and purification for scaled enzyme assays, an 8-l culture of *E. coli* BL21(DE3) pHis8-AhyBURP was expressed.

For purification of the other BURP-domain constructs, lysate supernatants were each directly subjected to flash-column chromatography with 5 ml of NiNTA resin washed with deionized water and equilibrated with lysis buffer. NiNTA-bound protein was washed with 20–30 column volumes of wash buffer (50 mM Tris-HCl, pH 8, 500 mM sodium chloride, 20 mM imidazole, 10 mM β -mercaptoethanol, 10% glycerol) and subsequently eluted with 10 ml of elution buffer (50 mM Tris-HCl, pH 8, 500 mM sodium chloride, 250 mM imidazole, 10 mM β -mercaptoethanol, 10% glycerol). For AhyBURP and CcaBURP1-[1xQLFW] purification, the 8xHis-tag and 6xHis-MBP tag was removed by TEV protease cleavage, respectively. Herein, TEV protease was added in a 1:50-molar ratio to the protein eluate and equilibrated in 3.5-kDa MWCO dialysis tubing against 2 l of 50 mM Tris-HCl, pH 8.0, 100 mM sodium chloride, 2 mM dithiothreitol at 4°C for 16 h. Cleaved protein tags were removed by flash-column chromatography of the dialyzed protein solution with 1 ml washed and dialysis buffer-equilibrated NiNTA resin. For CcaBURP2-[1xQLLVW], SkrBURP-[1xVLFYPSY] purification, eluate of 6xHis-MBP-tagged protein was equilibrated in 3.5-kDa MWCO dialysis tubing against 2 l of 50 mM Tris-HCl, pH 8.0, 100 mM sodium chloride, 2 mM dithiothreitol at 4°C for 16 h. Dialyzed protein solutions were concentrated in Sartorius Vivaspin concentrators with 10 kDa MWCO, analyzed by SDS-PAGE and used for enzyme assays or stored at -80°C.

In vitro BURP-domain assays. Enzyme reactions (50 μ l) were set up as follows with one metal cofactor per reaction. Metal cofactor screens included: (1) ferrous sulfate, (2) zinc chloride, (3) cupric chloride (all JT Baker Chemical Co.), (4) nickel(II) chloride hexahydrate or (5) cobalt(II) chloride hexahydrate (both Alfa Aesar). Enzyme assays for core peptide analysis included cupric chloride or water. AhyBURP assays: 90 μ M AhyBURP, 1 mM metal cofactor, citrate:phosphate buffer (citric acid:Na₂HPO₄, pH 5, 6 or 7, 35 μ l). CcaBURP2-[1xQLLVW] assays: 70 μ M MBP-CcaBURP2-[1xQLLVW], 1 mM metal cofactor, citrate:phosphate buffer (pH 5, 6 or 7, 35 μ l). CcaBURP1-[1xQLFW] assays: 20 μ M CcaBURP1-[1xQLFW], 1 mM metal cofactor, citrate:phosphate buffer (pH 5, 6 or 7, 35 μ l). SkrBURP-[1xVLFYPSY] assays: 50 μ M MBP-SkrBURP-[1xVLFYPSY], 1 mM metal cofactor, citrate:phosphate buffer (pH 5, 6 or 7, 35 μ l). Metal cofactor screens for SkrBURP-[1xVLFYPSY], CcaBURP2-[1xQLLVW] and CcaBURP1-[1xQLFW] were only set up at pH 7. Enzyme reactions were incubated for 24 h at 25°C, then 0.5 μ g trypsin (Sigma-Aldrich, T7575-1KT) dissolved in 20 μ l of 1 M Tris-HCl (pH 8) was added and incubated at 37°C for 16 h. Subsequently, tryptic digests were centrifuged, added to Chromacol vials and analyzed by LC-MS/MS for tryptic core peptides. Scaled enzyme assays of AhyBURP, SkrBURP-

[1xVLFYPSY] and CcaBURP1-[1xQLFW] (5 ml) for exoproteolytic cyclic peptide formation were based on purified BURP domains from 8-, 2- and 2-l expression cultures, respectively. Scaled enzyme assays included 20–50 μ M BURP domain, 1 mM CuCl₂ in citrate:phosphate buffer (pH 7). Enzyme assays were incubated for 24 h at room temperature. Subsequently, trypsin (Worthington Inc.) was added to the reaction mixtures in a 1:100 protease-substrate molar ratio after adjusting the pH to 8 with 1 M Tris-HCl (pH 8) buffer. Tryptic digests were incubated at 37°C for 24 h. Next, tryptic digests were subjected to preparative HPLC analysis (LC settings, Phenomenex Kinetex 5 μ m C18 100 Å LC Column 150 $\times$ 21.2 mm, 7.5 ml min⁻¹; solvent A, 0.1% TFA; solvent B, acetonitrile (0.1% TFA); LC gradient, 0 min, 10% B; 1 min, 10% B; 36 min, 50% B; 39 min, 95% B; 42 min, 95% B; 42.5 min, 10% B; 60.1 min, 10% B). HPLC fractions were analyzed by LC-MS analysis for target masses of modified core peptides with the following LC-MS parameters: injection volume 2 μ l; LC, Phenomenex Kinetex 2.6 μ m C18 reverse phase 100 Å 50 $\times$ 3 mm LC column; LC gradient, solvent A, 0.1% formic acid; solvent B, acetonitrile (0.1% formic acid); 0 min, 5% B; 2.5 min, 95% B; 3.0 min, 95% B; 3.1 min, 5% B; 5.0 min, 5% B, 0.5 ml min⁻¹; MS, positive ion mode; full MS, resolution 35,000, mass range 400–1,250 m/z; dd-MS², resolution 17,500; loop count 5, collision energy 25 eV and dynamic exclusion 0.5 s. LC fractions with target peptides were dried in vacuo and resuspended in 30 μ l of DMSO. Exopeptidase assays were performed with approximately 25–50 μ g peptides (4% [reaction DMSO]), carboxypeptidase Y (*S. cerevisiae*, Sigma-Aldrich) and aminopeptidase N (Rat, Sigma-Aldrich) in 1:10 (w/w) ratio of peptidase:peptides, in 100 mM Tris-HCl (pH 7) and 1 mM EDTA at 37°C or 24 h. The exopeptidase assays were centrifuged for 5 min at 18,000g and analyzed by LC-MS in comparison to purified natural products (AhyBURP, legumenin; CcaBURP1-[1xQLFW], cercic acid) or *N. benthamiana* gene expression extract (SkrBURP-[1xVLFYPSY], pEAQ-HT-SkrBURP) as described for tryptic digests of enzyme assays in the Proteomics section with adjusted mass ranges to include target cyclic peptide products as follows: AhyBURP, 700–1,000 m/z; SkrBURP, 500–1,400 m/z and CcaBURP1, 600–1,000 m/z.

Proteomics. Tryptic digests of enzyme assays were analyzed by LC-MS/MS with the following common settings for all analyses: injection volume 15 μ l; LC, Higgins Analytical PROTO300 C4 5 μ m 250 $\times$ 4.6 mm; all gradients: solvent A, 0.1% formic acid; solvent B, acetonitrile (0.1% formic acid); 0.7 ml min⁻¹; MS, positive ion mode. Metal cofactor screens were analyzed as follows: LC gradient 0 min, 10% B; 10 min, 50% B; 11 min, 95% B; 13 min, 95% B; 13.1 min, 10% B; 20 min, 10% B; MS, full MS resolution 70,000, mass range 700–1,000 m/z (AhyBURP, CcaBURP1-[1xQLFW], CcaBURP2-[1xQLLVW]) or 1,250–1,400 m/z (SkrBURP-[1xVLFYPSY]), dd-MS²: resolution 17,500, AGC target 1 $\times$ 10⁵, loop count 5, isolation width 0.4 m/z, collision energy 25 eV and dynamic exclusion 0.5 s. AhyBURP enzyme assays were analyzed as follows: LC gradient, 0 min, 15% B; 60 min, 40% B; 61 min, 95% B; 63 min, 95% B; 63.1 min, 15% B; 80 min, 15% B; MS, full MS resolution 70,000, mass range 1,250–1,400 m/z; dd-MS², resolution 17,500, AGC target 1 $\times$ 10⁵, loop count 5, isolation width 0.4 m/z, collision energy 25 eV and dynamic exclusion 0.5 s. MS settings for core peptide fragmentation analyses were as follows: (1) linear core peptide, targetedSIM, target mass 813.9194 m/z (z = 2), MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV; MS² isolation width 0.4 m/z, AGC target 1 $\times$ 10⁵; (2) monocyclic core peptide, targetedSIM, target mass 812.9122 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV; MS² isolation width 0.4 m/z; (3) bicyclic core peptide, targetedSIM, target mass 811.9049 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV; MS² isolation width 0.4 m/z; AGC 1 $\times$ 10⁵. SkrBURP-[1xVLFYPSY] enzyme assays were analyzed as follows: LC gradient, 0 min, 15% B; 60 min, 40% B; 61 min, 95% B; 63 min, 95% B; 63.1 min, 15% B; 80 min, 15% B; MS, full MS, resolution 70,000; mass range, 1,250–1,400 m/z; dd-MS², resolution 17,500; AGC target 1 $\times$ 10⁵; loop count 5, isolation width 0.4 m/z, collision energy 20 eV and dynamic exclusion 0.5 s. MS settings for core peptide fragmentation analyses were as follows: (1) linear core peptide, targetedSIM, target mass 1,377.1563 m/z (z = 2), MS¹ isolation width 4 m/z; dd-MS², HCD 20 eV, MS² isolation width 0.4 m/z, AGC target 1 $\times$ 10⁵; (2) monocyclic core peptide, targetedSIM, target mass 1,376.1485 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 20 eV, MS² isolation width 0.4 m/z; (3) bicyclic core peptide, targetedSIM, target mass 1,375.1407 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 20 eV; MS² isolation width 0.4 m/z and AGC 1 $\times$ 10⁵. CcaBURP1-[1xQLFW] enzyme assays were analyzed as follows: LC gradient, 0 min, 10% B; 10 min, 50% B; 11 min, 95% B; 13 min, 95% B; 13.1 min, 10% B; 20 min, 10% B. MS settings for core peptide fragmentation analyses were as follows: (1) linear core peptide, targetedSIM, target mass 869.4334 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV, isolation width 0.4 m/z and AGC target 1 $\times$ 10⁵; (2) monocyclic core peptide, targetedSIM, target mass 868.4256 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV; MS² isolation width 0.4 m/z, CcaBURP2-[1xQLLVW] enzyme assays were analyzed as follows: LC gradient, 0 min, 10% B; 10 min, 50% B; 11 min, 95% B; 13 min, 95% B; 13.1 min, 10% B; 20 min, 10% B. MS settings for core peptide fragmentation analyses were as follows: (1) linear core peptide, targetedSIM, target mass 876.4265 m/z (z = 2); MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV, MS² isolation width 0.4 m/z and AGC target 1 $\times$ 10⁵; (2) monocyclic core peptide, targetedSIM, target mass 876.4187 m/z (z = 2), MS¹ isolation width 4 m/z; dd-MS², HCD 25 eV and MS² isolation width 0.4 m/z.

Linear and modified core peptide fragments were analyzed manually in Qualbrowser. MS/MS analysis of putative monocyclic AhyBURP tryptic core peptide (IPFAQPYGVYTWGK, parent mass 812.9115 *m/z*, 2+) showed b12–b14 ion masses with loss of two hydrogens and y7–y13 ion masses with loss of two hydrogens (Supplementary Fig. 9) were detected indicating a monocyclization modification in the sequence GVVYTW. In addition, unmodified b7 and b2–b5 ions and unmodified y1 and y2 ions were detected supporting a lyciumin-type monocyclization in this tryptic core peptide. The detection of unmodified y3 and y4 ions is most likely due to the lability of the C–N-macrocyclic bond between the tryptophan-indole-nitrogen and the C α of the glycine. MS/MS analysis of the putative bicyclic tryptic core peptide (parent mass 811.9049 *m/z*, 2+) showed b12–b14 ions with loss of four hydrogens, y10–y14 ions with loss of four hydrogens, b2–b4 without loss of hydrogens and y1 and y2 without loss of hydrogens. These detected ions supported a bicyclic modification within the sequence of QPYGVYTW. The detected y9 ion with a loss of two hydrogens and the y3 and y4 ions without mass losses might be due to the lability of the proposed macrocyclic C–N- and C–O bonds. MS/MS analysis of putative monocyclic SkrBURP-[1xVLFYPSY] tryptic core peptide (AVPASVEDPSQADADNVLFYPSYAAR, 1,376.1489 *m/z*, 2+) showed y9–y26 ions with loss of two hydrogens, b20–b26 with loss of two hydrogens, y2–y6 ions without mass loss and b3–b17 ions without mass loss indicating a monocyclic modification in the sequence LFY (Supplementary Fig. 10). The detection of unmodified y7 and y8 ions could be due to the labile nature of the leucine-tyrosine-macrocyclic ether bond. MS/MS analysis of putative bicyclic SkrBURP-[1xVLFYPSY] tryptic core peptide (AVPASVEDPSQADADNVLFYPSYAAR, 1,375.1416 *m/z*, 2+) showed y9–y23 ions and y26 ion with mass loss of four hydrogens, y2 and y3 ions without mass loss, b24–b26 ions with mass loss of four hydrogens, b3–b8 and b10–b17 ions without mass loss indicating a bicyclic modification in the sequence LFYPSYA. The detection of the y8 ion with mass loss of two hydrogens and y6 ion without mass loss could be due to the labile nature of the macrocyclic ether-bonds. MS/MS analysis of putative monocyclic CcaBURP2-[1xQLLVW] tryptic core peptide (TICDDLNSQLLVWR, 875.4345 *m/z*, 2+) showed y5–y12 and y14 ions with mass loss of two hydrogens, b14 ion with mass loss of two hydrogens, b2–b9 ions without mass loss and y1 ion without mass loss indicating a monocyclic modification in the sequence LLVWR (Supplementary Fig. 11). MS/MS analysis of putative monocyclic CcaBURP1-[1xQLLFW] tryptic core peptide (AIYDDPNSQLFWR, 869.4335 *m/z*, 2+) showed y5–y12 ions with mass loss of two hydrogens, b13 and b14 ions with mass loss of two hydrogens, b2–b5 and b7–b9 ions without mass loss and y1 ion without mass loss indicating a monocyclic modification in the sequence ILFW (Supplementary Fig. 12).

RepeatFinder development and plantISMASH integration. The RepeatFinder analyzes the protein sequences from open reading frames near a BURP domain detected by plantISMASH. First, it indexes all occurring *k*-mers and ranks them by frequency of occurrence *f*. The most frequent *k*-mer becomes a basis for a table of *k* columns and *f* rows. New columns are appended to the table in iterations, in which each residue in the column is the next residue downstream of the *k*-mer in that row. The formula for the position of these residues in the original sequence is $kpos + k + i$, in which *kpos* is the position of the first residue of the *k*-mer and *i* is the iteration index. These newly added columns are scored by calculating the BLOSUM (Blocks Substitution Matrix) score⁶⁰ of each residue against the most frequently occurring residue in the column. These scores are averaged to score the similarity of the column. When the score of the column drops below a predefined value (2.5), or the pattern reaches a predefined maximum length (15 amino acids), the expansion is ended and the algorithm proceeds to the filtering step. The filtering step removes any row that does not contain more than two different residues. This dramatically cuts back on false positives among highly repetitive sequences and low complexity regions. The location of all repeat instances as well as the pattern is saved and shown as visual output below the cluster in the plantISMASH HTML output, which can be viewed in any modern browser by the user. In addition, known core peptide motifs are identified in open reading frames surrounding BURP domains and are displayed in the plantISMASH output. Known core peptide annotations were analyzed manually given described BURP-domain precursor-peptide features, that is N-terminal sequence repeats or core peptide within BURP domain (AhyBURP).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Plant genome sequences were derived from JGI Phytozome (v.13)⁴⁵ and NCBI GenBank. Known peptide structures were derived from Pubchem⁶². Gene sequences of peptide precursors have been deposited in GenBank (CcaBURP1, MW570736; CcaBURP2, MW570737; AhyBURP, MW570734 and SkrBURP, MW570735). Transcriptomic data of *S. kraussiana* has been deposited in the NCBI-SRA (SRR11647978). Metabolomic datasets and reference peptide MS/MS spectra have

been submitted to GNPS-MassIVE²⁹ under the accession number MSV000087872. MS/MS spectra of characterized peptides have been submitted to GNPS-MS/MS Spectral Libraries. The molecular network can be found under <https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=c530b123d3984aa58247d6bd2789c69f#>. Plant genome analysis outputs by plantISMASH-RepeatFinder are available at: <https://doi.org/10.5281/zenodo.4687169>. All other data supporting the findings of this study are presented in the published article including its supporting information or are available from the corresponding authors upon reasonable request.

Code availability

Updated plantISMASH and RepeatFinder code are available on Github (<https://github.com/plantismash/plantismash>).

References

- Kressner, D., Chambers, M., Burke, R., Agus, D. & Mallick, P. ProteoWizard: open source software for rapid proteomics tools development. *Bioinformatics* **24**, 2534–2536 (2008).
- Shannon, P. Cytoscape: A Software environment for integrated models of biomolecular interaction networks. *Genome Res.* **13**, 2498–2504 (2003).
- Bankovich, A. et al. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J. Comput. Biol.* **19**, 455–477 (2012).
- Altschul, S. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.* **25**, 3389–3402 (1997).
- Priyam, A. et al. Sequenceserver: a modern graphical user interface for custom BLAST databases. *Mol. Biol. Evol.* **36**, 2922–2924 (2019).
- Gibson, D. G. et al. Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat. Methods* **6**, 343–345 (2009).
- Madeira, F. et al. The EMBL-EBI search and sequence analysis tools APIs in 2019. *Nucleic Acids Res.* **47**, W636–W641 (2019).
- Frisch, M. J. et al. Gaussian 16 (Gaussian.com, 2016).
- Frederick, R. O. et al. Small-scale, semi-automated purification of eukaryotic proteins for structure determination. *J. Struct. Funct. Genomics* **8**, 153–166 (2007).
- Henikoff, S. & Henikoff, J. G. Amino acid substitution matrices from protein blocks. *Proc. Natl Acad. Sci. USA* **89**, 10915–10919 (1992).
- Kim, S. et al. PubChem substance and compound databases. *Nucleic Acids Res.* **44**, D1202–D1213 (2016).

Acknowledgements

We thank D. Michener and M. Palmer for access to the Matthaei Botanical Garden and C. Dick and B. Ruhfel for access to the University of Michigan Herbarium. We also thank G. Lomonosoff (John Innes Centre, UK) for sharing the pEAQ-HT vector. This work was supported by the Biological Sciences Scholar Program at the University of Michigan (R.D.K.) and the Graduate School for Experimental Plant Sciences in the Netherlands (M.H.M.).

Author contributions

D.N.C., L.S.M. and F.d.W. contributed equally to this work. R.D.K. and M.H.M. conceived the idea. D.N.C., W.L., K.S. and R.D.K. collected plant samples, prepared and analyzed plant extracts by metabolomics, isolated and structurally elucidated plant peptides. R.D.K. expressed plant proteins in tobacco, prepared and analyzed transgenic tobacco leaf extracts by metabolomics. L.S.M. and R.D.K. expressed proteins in *E. coli*, purified proteins, performed and analyzed in vitro enzyme assays by bottom-up proteomics. T.M. analyzed known plant cyclic peptide structures. K.S., J.W.W., O.G.M., A.T. and J.Z.S. performed peptide bioactivity assays. M.H.M., S.K. and F.d.W. designed and wrote RepeatFinder code and integrated RepeatFinder into plantISMASH. M.H.M., S.K., F.d.W. and R.D.K. analyzed plant genomes via plantISMASH-RepeatFinder. R.D.K., M.H.M., D.N.C., L.S.M. and F.d.W. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

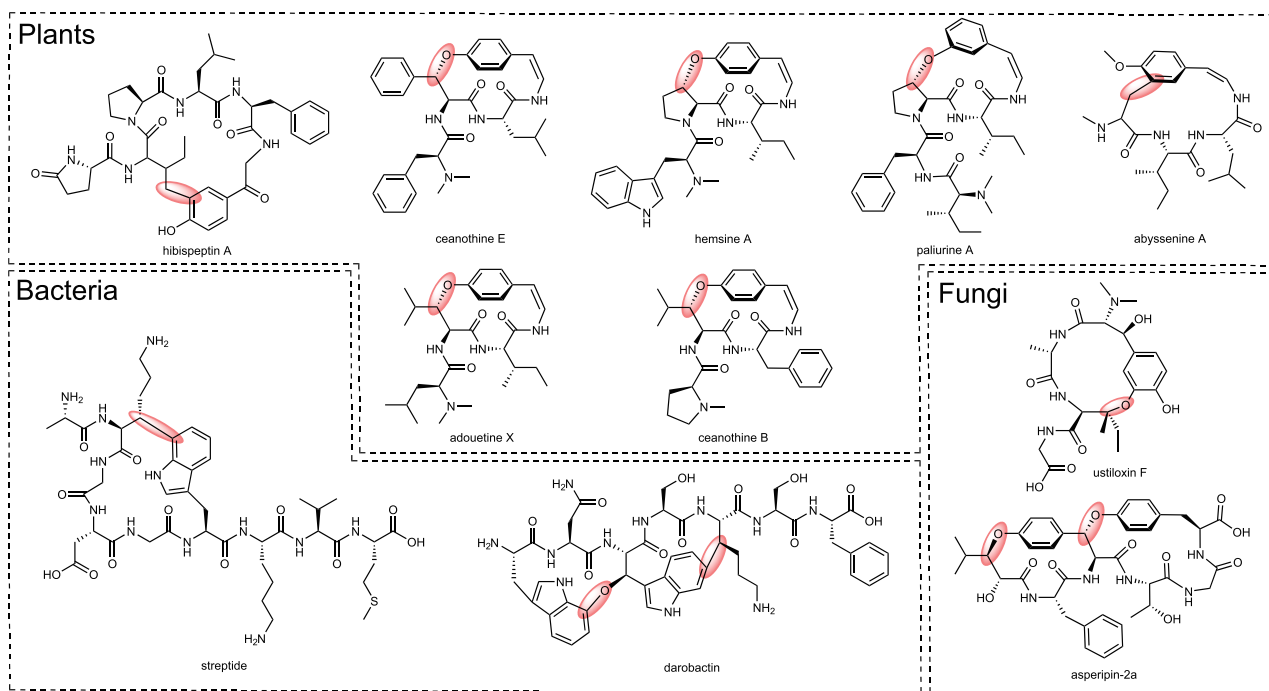
Extended data are available for this paper at <https://doi.org/10.1038/s41589-021-00892-6>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41589-021-00892-6>.

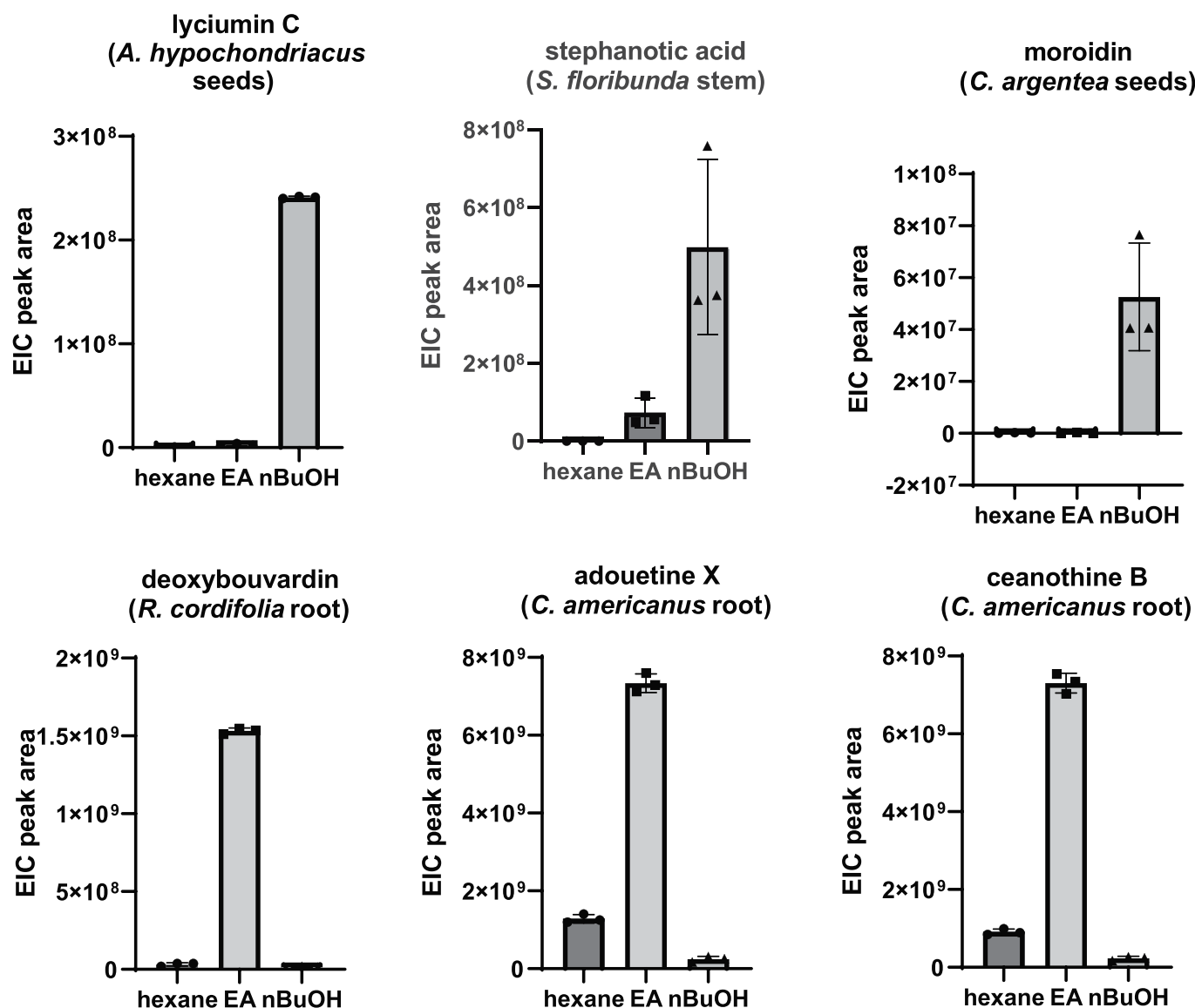
Correspondence and requests for materials should be addressed to Marnix H. Medema or Roland D. Kersten.

Peer review information *Nature Chemical Biology* thanks Yit-Heng Chooi and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

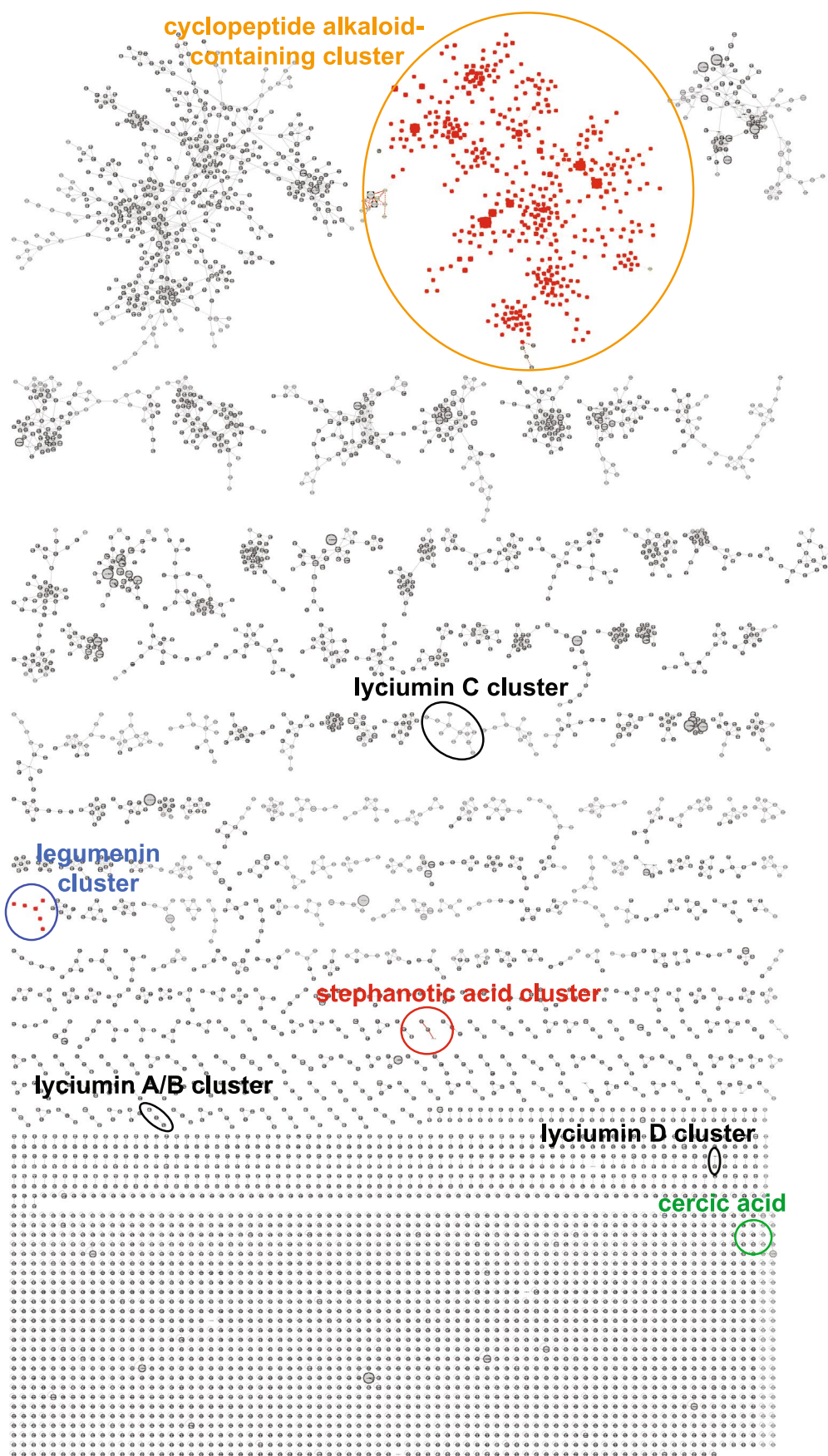
Reprints and permissions information is available at www.nature.com/reprints.



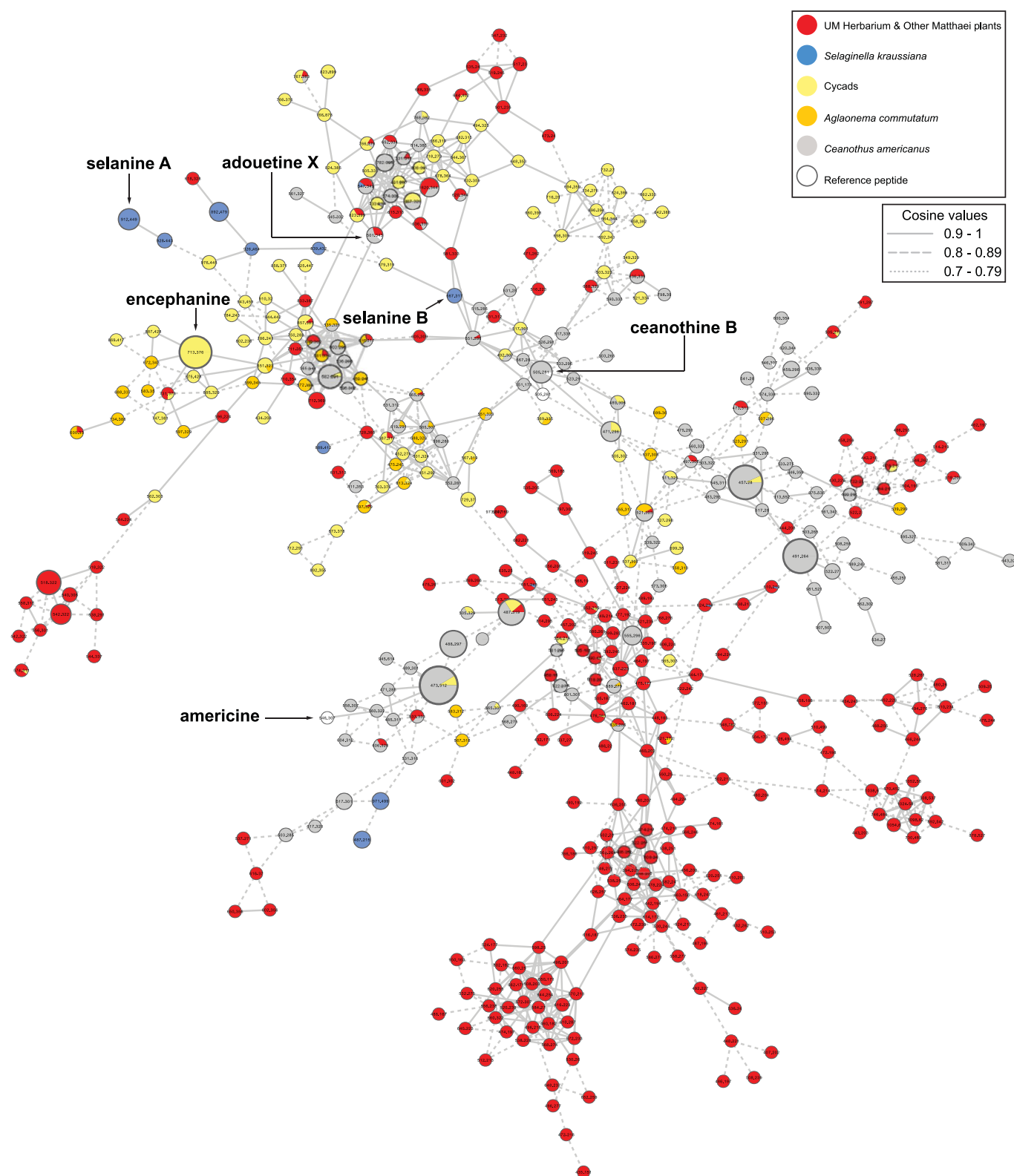
Extended Data Fig. 1 | Peptides with side-chain-macrocyclizations from plants, bacteria and fungi. Representative chemical structures are shown. Macrocyclic bonds are highlighted in red.



Extended Data Fig. 2 | Extraction analysis of reference plant peptides with side-chain-macrocyclic bonds from source plant material. Reference peptides were monitored in extract fractions based on corresponding extracted ion chromatogram (EIC) peak areas. EIC peak area data were plotted as dot plots overlayed with bar graphs representing the mean of three extractions of tissue from three biologically different plants ($N=3$). Error bars of the bar graph represent one standard deviation ($\pm$ SD). Abbreviation: EA - ethyl acetate, nBuOH - *n*-butanol, *A. hypochondriacus* - *Amaranthus hypochondriacus*, *S. floribunda* - *Stephanotis floribunda*, *C. argentea* - *Celosia argentea*, *R. cordifolia* - *Rubia cordifolia*, *C. americanus*.



Extended Data Fig. 3 | Molecular network of peptide-filtered LC-MS/MS datasets of UM botanical collections. Full molecular network of peptide-filtered datasets with characterized spectral clusters highlighted including new cyclic plant peptides.



Extended Data Fig. 4 | Cyclopeptide alkaloid-containing spectral cluster of peptide-filtered LC-MS/MS datasets of UM botanical collections. Nodes are colored according to source plant as shown in the top box. Node sizes represent relative precursor ion intensity. Node values are precursor ion mass values. Pie charts represent relative spectral counts of defined groups. Nodes of reference peptides and characterized peptides are indicated with arrows.

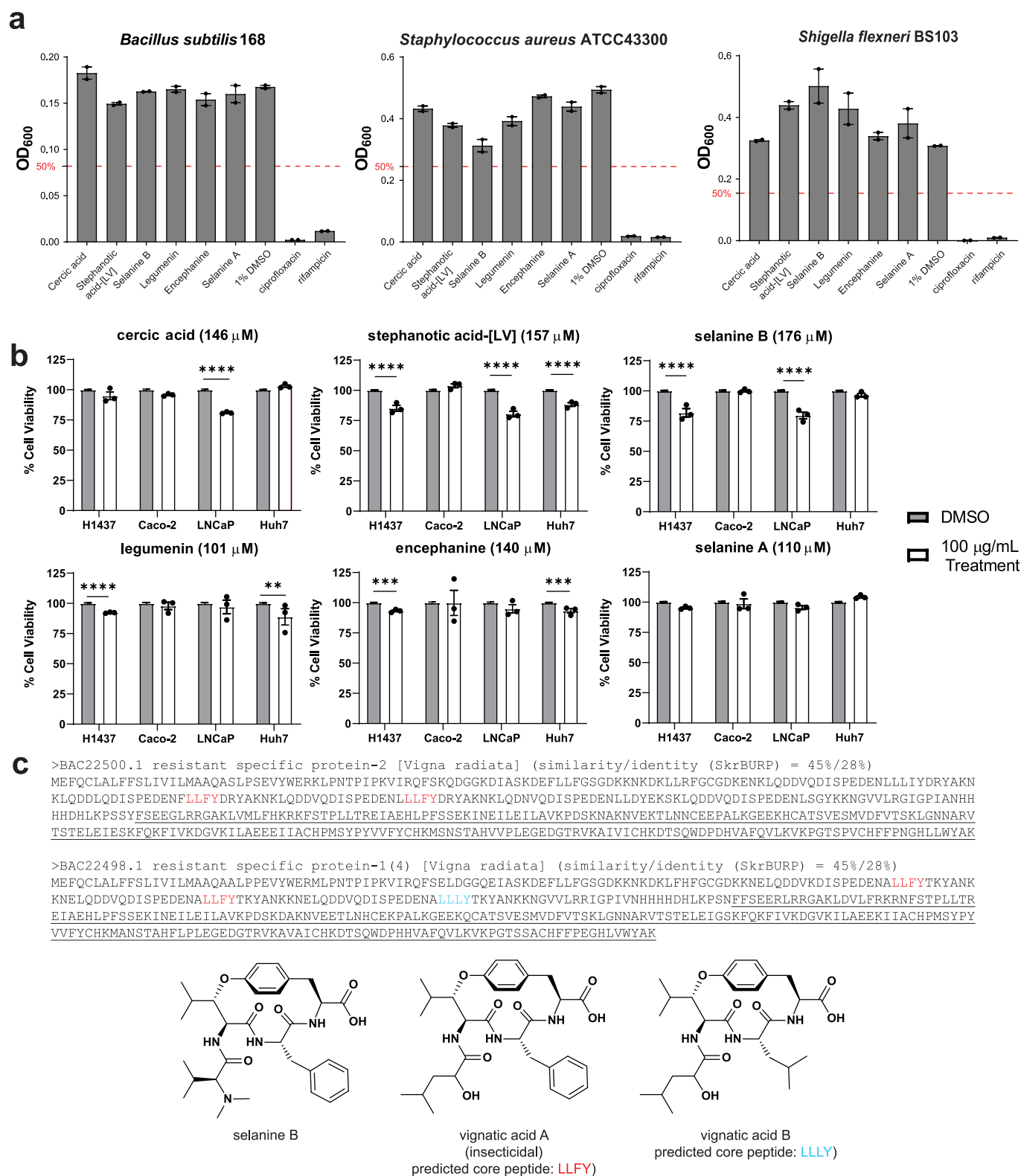
>SkrBURP (*Selaginella kraussiana*, precursor for 1 & 2)
 MAQSLILLFLIVMGCYDVGAIERVHSAKEGVSE
 AMNDQGHPTIANKAVPASVEDPSQADADNILLY
 PSYAACKAVPEDPSQADADNVLFYRSYAACKAV
 PASVEDPSHDADNVLFYPSYVAKKAVPASVEDP
 SQADADNFFLYPYAAKAVPASVEDPSHDADNV
 LFYPSYVAKKAVPASVEDPSQADADNLFYPYA
 AKKAVPASVEDPSHDADNVLFYPSYVAKKAVPA
 SVEDPSHDADNVLFYPSYVAKKAVPASVEDPSQ
 ADADNVLFYPSYAARPKAPSTMDDHATHNAAQN
 MHHQKALLFFRMNTLKVGGEVVIPSLASHALG
 GRKLMTPRLEAMLGSMELPRLLSVLKIPLDSTL
 ARKSKTNLGNCRSPPLQGEKKACVSSIRSMTNF
 ARSVLEDKPLEHLNPSRPVSSPEHFKIMDVKV
 VTENSVVTCHEPMVFPYALYMCHFPVKSVPKVT
 LQDDHDKLVVVPVMCHMDTSEFDPShLSFKILN
 TKPGEAEMCHWMPNSHIMWYTSDGKTRDVL

>CcaBURP2 (*Cercis canadensis*, precursor for 4)
 MKPHRLPMFIFLIFAPVVINAALPAELYW
 KSMLPNTTMPKAIKDLLSAGNMGTICDD
 LNSQLLVWQLHEDPAAIDQLRDDPNSQLLVWQKHEDPAATDQLHDDPNSQLLVWQKHE
 GPAAIDQLRDDPNSQLKVWQEHKHPVATD
 QLRDDPNSQLKVWQKHEDPTATDQLRDDP
 NSQLKVWQKHEDPEAIDQLRDDPNSQLLVWQKHEDPAATDQLRDDPNSQLLVWRHCAT
 ETHSRDSSGKFLFSENDLHPGKKLKMFL
 TKSYPYPTLIPHEVADTIPFSLKKVPEIL
 KMFSIEPESKKAKIIGVHHCEPPTKG
 EEKHCATSLISMVDYATSKLGKNVKAYST
 EVEKETKLQTYTIAPGVKKISDEVMAHQ
 THYPYVVFHCHQVGLSRAYFVPLEGSDG
 TRVKAIVVCHSDTSDWDVDHLAFRTLKID
 PGTASVCHFSPADNVVVIHKTT

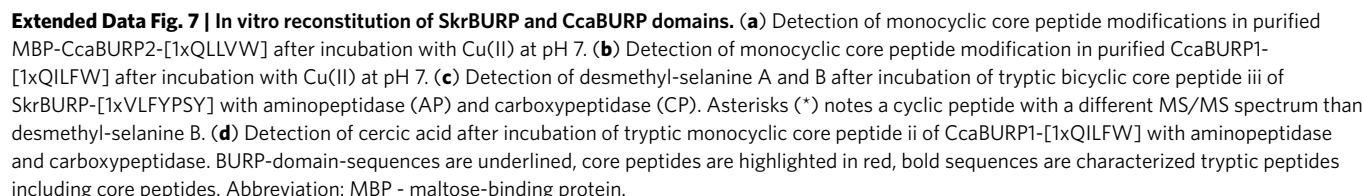
>CcaBURP1 (*Cercis canadensis*, precursor for 5)
 MKLHRLTMFIFLILAPVVINAALPPELYWKSMLPNTTMPKA
 VKDLLSAGNMEKAIYDDPNSQILFWRHAATSADQLHDDPNT
 QILFWRHAATTIDQLHDDPNSQILFWRRVATSTDQLHKDPN
 SQILFWRRATSADQLHDDPNSQILFWRSATTTVDQLHDDPN
 SQILFWRHAATSADQLHKDPNSQILFWRRATSADQLHDDPN
 SQILFWRSATTTVDQLHDDPNSQILFWRNAATSADQLHKDP
 NSQILFWRRASTAANQLHDDPNAQILFWRHAATTADQLHDD
 SNTQILFWRCATTTTQDLHDDPNSQILFWRLAATAVDQLH
 DDPNGQILFWRHATTATTEQLHNDPNAQILFWRHADQLHND
 PNTQILFWRHATPMADQLHDDPNSQILFWRRSAIEAHSRDD
 SCKFMFFSENDLHPGKKLEMLFTKSPYSTALIPEVADKIP
 FSLKKVPEILKMFSIEPESKEAKIIGAIHHCQEPPTKGE
 EKHCAATSLISMVDYVNSKLGKNVRAYSTEVEKETKLQTYTI
 APGVKKISDEVMAHQTHYPYVLFHCHQGGGSTRAYFVPLE
 GSDGTRVKAIVCHSDTSDWDVDHLAFRTLKIERGTAPVCH
 FSPADNVITWPQKTT

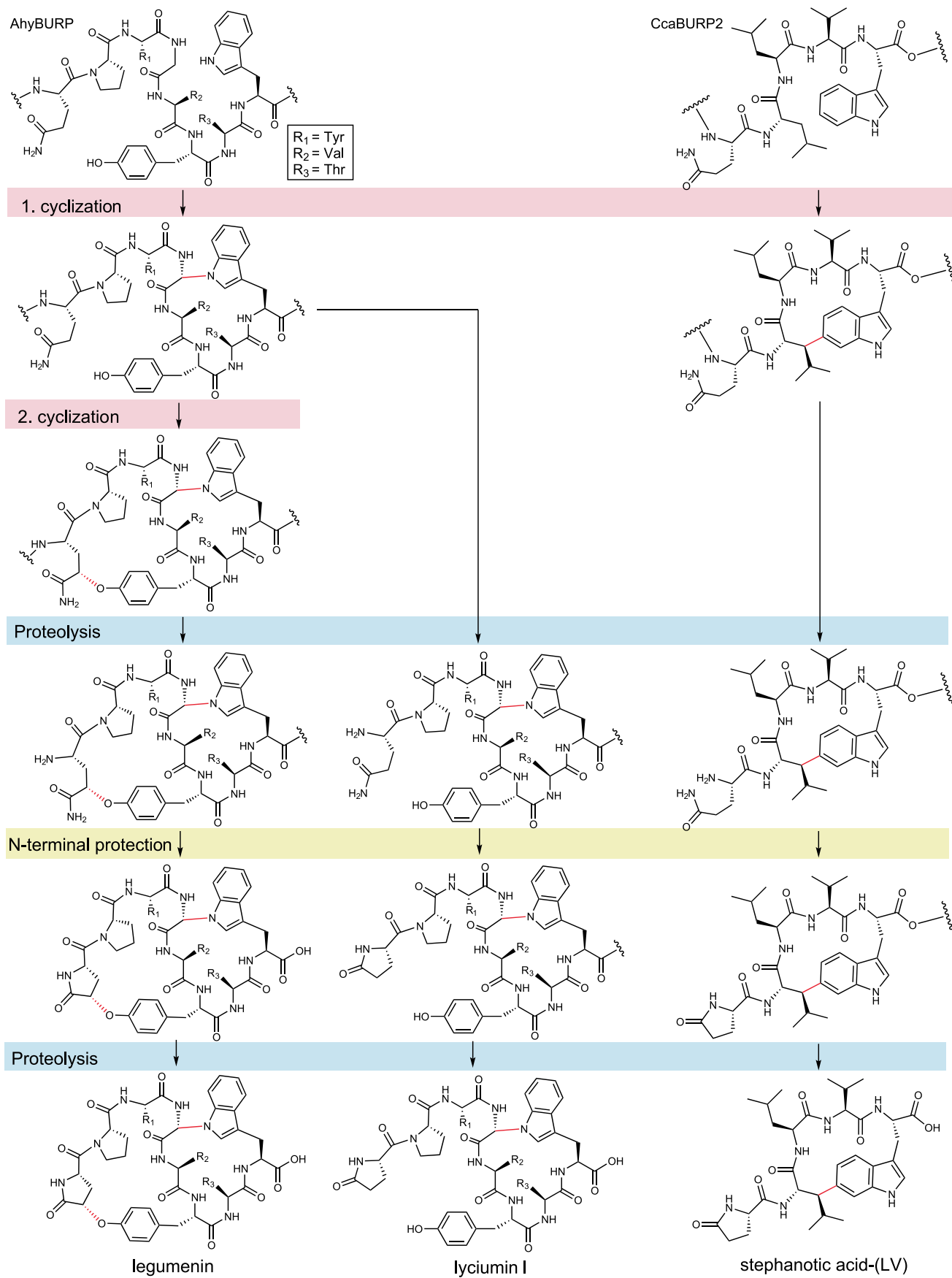
>AhyBURP (*Arachis hypogaea*, precursor for 6)
 MEFRCAVAMLFCLAFAGSRG
 RELIPDEDYQAVWPNTPIPN
 TLKELLKPGAQDSEINDVPMK
 VDDTQYPKTTFFFEHELFPGKK
 MNMKFSKIPFAQPYGVYTWGK
 VIKDLEKESFTFEDACVREAG
 KGEDKYCAKSLSTLIGFAVSK
 LGKNIQPFSSSFLDKQTDYTI
 EGVHNLGDKAVMCHRLNFQST
 VFYCHEIHGTTAYMVPMVAAD
 GRRTQALAVCHHDTSGMNAEV
 LYEMLKIKPGTETACHFLGNK
 AVMWVPMNAVNSVYNNANMAS

Extended Data Fig. 5 | Candidate BURP domain precursor peptides of target cyclic plant peptides from botanical collections. BURP domains are underlined, putative core peptides are highlighted in red.



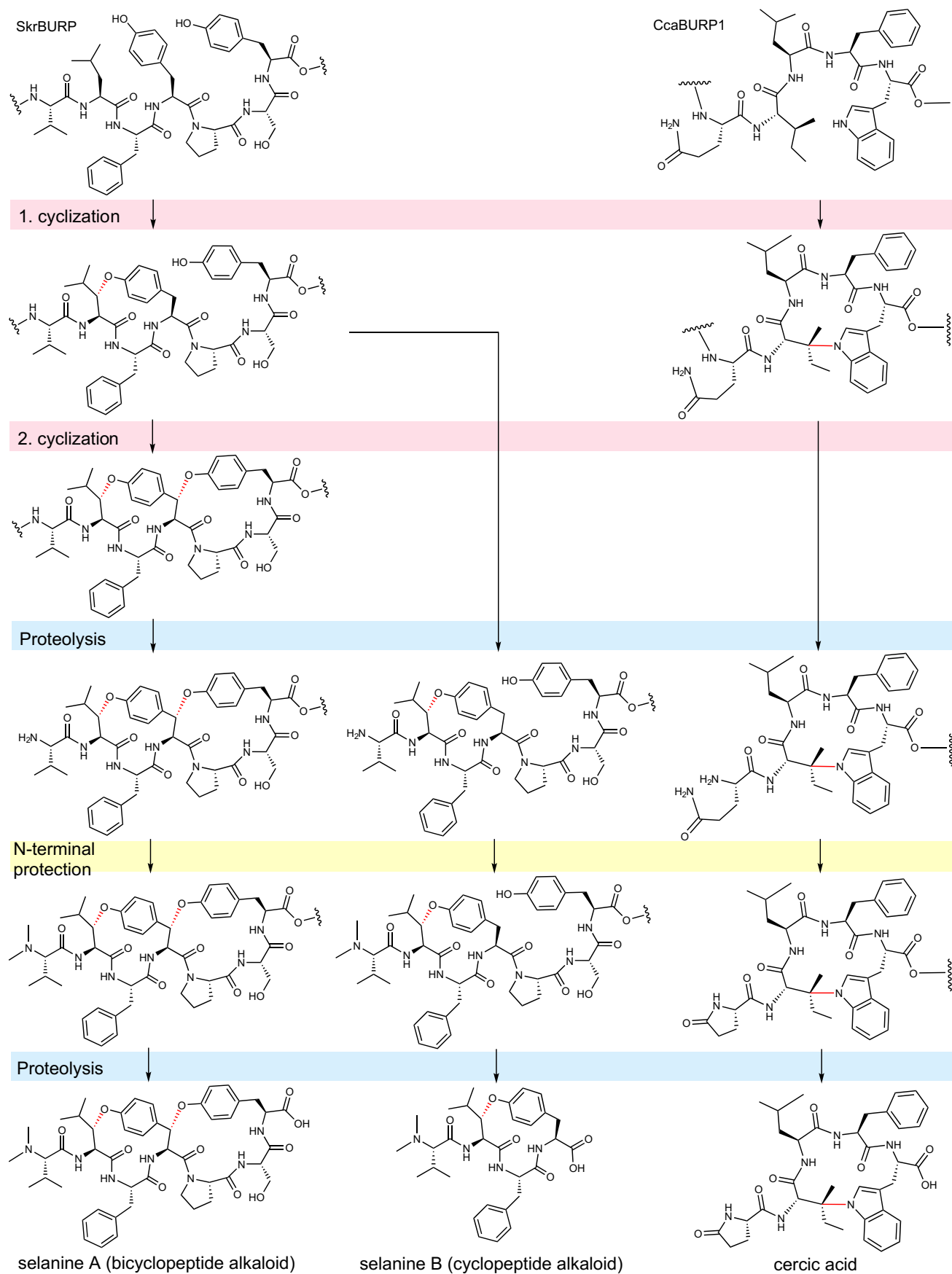
Extended Data Fig. 6 | Bioactivity evaluation of side-chain-macrocytic plant peptides. (a) Antibacterial assays show no significant growth inhibition of *Bacillus subtilis* 168, *Staphylococcus aureus* ATCC43300 or *Shigella flexneri* BS103 at 100 μ M compound concentration compared to 1% DMSO control after 24 h incubation at 37 °C compared to antibiotic positive controls of ciprofloxacin and rifampicin. Data represents mean of OD600 of cultures ($N=2$) $\pm$ SEM. (b) Compounds were assessed for anti-cancer activity at a concentration of 100 μ g/mL in four different human cell lines including H1437 (lung adenocarcinoma), Caco-2 (colorectal adenocarcinoma), LNCaP (prostate adenocarcinoma), and Huh7 (hepatocellular carcinoma). Data represents the mean $\pm$ SEM of $N=68$ vehicle controls and $N=3$ treatments. Statistically significant reductions in cell viability were determined using unpaired two-tailed t -tests in Graphpad Prism 9 ($p < 0.0001$ = ****, $p < 0.001$ = ***, $p < 0.01$ = **). (c) Sequences of BURP domain protein homologs of selanine precursor peptide SkrBURP. SkrBURP homologs from bruchid-resistant *Vigna radiata* contain core peptides which match the insecticidal cyclopeptide alkaloid vignatic acid A from bruchid-resistant *Vigna radiata* strain TC1966.





Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Biosynthetic proposal for BURP-domain-derived plant RiPPs legumenin, lyciumin and stephanotic acid-[LV]. Biosynthesis of BURP-domain-derived plant RiPPs is proposed to proceed through BURP domain cyclization of core peptide motifs, N-terminal proteolysis, N-terminal protection and C-terminal proteolysis. Legumenin biosynthesis involves two sequential cyclizations, in which the C-terminal macrocycle is being formed first. Formed macrocyclic bonds in core peptide motifs are highlighted in red.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Biosynthetic proposal for BURP-domain-derived plant RiPPs selanine A (bicyclopeptide alkaloid), selanine B (cyclopeptide alkaloid) and cercic acid. Biosynthesis of BURP-domain-derived plant RiPPs is proposed to proceed through BURP domain cyclization of core peptide motifs, N-terminal proteolysis, N-terminal protection and C-terminal proteolysis. Selanine A biosynthesis involves two sequential cyclizations, in which the N-terminal macrocycle is formed first. Formed macrocyclic bonds in core peptide motifs are highlighted in red.

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

Mass spectrometry data was collected with Thermo Xcalibur version 4.3.73.11. NMR data analysis was done with Bruker Topspin (version 4.0.8). Antibiotic activity data was collected with SoftMax Pro Software (Molecular Devices LLC). Anticancer activity data was collected by source image analysis software Cellprofiler.

Data analysis

Mass spectrometry data was analyzed with Qualbrowser in the Thermo Xcalibur software suite. Molecular networking analysis was done with GNPS-molecular networking (release 26) (<https://gnps.ucsd.edu/ProteoSAFe/static/gnps-splash.jsp>) and molecular networking data was analyzed with Cytoscape (Version 3.8). NMR data analysis was done with Bruker Topspin (version 4.0.8). Antibiotic bioactivity data, source plant reference peptide quantification and anti-cancer bioactivity data was analyzed by GraphPrism 9 (GraphPad Software Inc., La Jolla, CA). Genomic data was analyzed by plantiSMASH (<http://plantismash.secondarymetabolites.org/>) with a software extension RepeatFinder. The corresponding code is deposited on GitHub at <https://github.com/plantismash/plantismash>. BLAST searches were done with blastp on an internal Blastserver or on NCBI website (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). De novo RNA-seq data assembly was done by SPAdes (version 3.14.0) (<https://cab.spbu.ru/software/spades/>). Molecular geometry optimization was done with Chem3D (version 18.1, PerkinElmer) and Gaussian (version 16 A.03, Gaussian inc.).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Plant genome sequences were derived from JGI Phytozome (version 13) and NCBI GenBank. Gene sequences of peptide precursors have been deposited in GenBank (CcaBURP1: MW570736, CcaBURP2: MW570737, AhyBURP: MW570734, SkrBURP: MW570735). Transcriptomic data of *Selaginella kraussiana* has been deposited in the NCBI-SRA (SRR11647978). Metabolomic datasets and reference peptide MS/MS spectra have been submitted to GNPS-MassIVE under the accession number MSV000087872. MS/MS spectra of characterized peptides have been submitted to GNPS-MS/MS Spectral Libraries. The molecular network can be found under: <https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=c530b123d3984aa58247d6bd2789c69f#>. Plant genome analysis outputs by plantiSMASH/RepeatFinder are available at: <https://doi.org/10.5281/zenodo.4687169>. All other data supporting the findings of this study are presented in the published article including its Supporting information or are available from the corresponding authors upon reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculations were performed. The sample size for detection of reference peptides in different solvent partitioning fractions from source plants was $n = 3$ as a minimum number to determine the detection of differentially extracted natural products (Extended Data Fig. 2). The sample size for cancer cell cytotoxicity assays was $n=3$ for compound treatments to enable statistical testing of cell viability inhibition compared to the control ($n=68$). The sample size for assays to test antibacterial growth inhibition of compounds 1-6 was $n=2$ as no significant growth inhibition (i.e. less than 50% growth inhibition) at high compound concentrations was detected and the errors were small (Extended Data Fig. 5).
Data exclusions	No data was excluded from experiments.
Replication	Transient expression experiments in <i>N. benthamiana</i> and source plant extractions of target compounds were repeated successfully in triplicate from biologically different plants (Figure 2, 3 & 6). Enzyme assays (Fig. 4, 5 & Extended Data Fig. 6) were reproducible in triplicate. Protein purification from <i>E. coli</i> was reproducible in triplicate.
Randomization	Compound treatments of cancer cells in anti-cancer assays were block randomized. <i>N. benthamiana</i> plants with injected target genes and injected empty vector controls were randomly positioned under growth lights.
Blinding	Antibacterial and anti-cancer assays were blinded in the treatment of target compounds 1-6.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	Cell lines Huh7, H1437, Caco2 and LnCAP were purchased from ATCC within 3 years.
Authentication	The cell lines were authenticated by ATCC. Seventeen short tandem repeat (STR) loci plus the gender determining locus, Amelogenin, were amplified using the commercially available PowerPlex® 18D Kit from Promega. The cell line samples were processed using the ABI Prism® 3500xl Genetic Analyzer. Data were analyzed using GeneMapper® ID-X v1.2 software (Applied Biosystems). Appropriate positive and negative controls were run and confirmed for each sample submitted.
Mycoplasma contamination	Cells were tested for mycoplasma contamination on May 27th, 2021 using the MycoAlert™ Mycoplasma Detection Kit (Lonza LT07-318) and all were negative (Huh7, H1437, CaCo2, LNCaP).
Commonly misidentified lines (See ICLAC register)	n/a